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Zhang et al.

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(54) **HERBICIDE-RESISTANT ACETYL-CoA
CARBOXYLASE (ACC) MUTANT AND USE
THEREOF**

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C12N 9/00 (2006.01)

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(2013.01); **C12N 15/8213** (2013.01); **C12Y**
604/01002 (2013.01)

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None
See application file for complete search history.

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(57) ABSTRACT

A mutant acetyl-CoA carboxylase (ACC) protein, a nucleic
acid encoding the mutant ACC protein, and use thereof are
provided. Specifically, compared with a parent ACC protein,
the mutant ACC protein has mutations at amino acids
corresponding to amino acid 1,879 and/or amino acid 2,186
of SEQ ID NO: 1. An ACC-mutated plant shows high
herbicide resistance, and thus the present disclosure has very
promising application prospects in the cultivation of an
herbicide-resistant plant.

8 Claims, 3 Drawing Sheets

Specification includes a Sequence Listing.

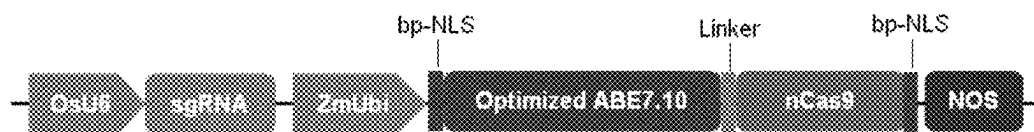


FIG. 1

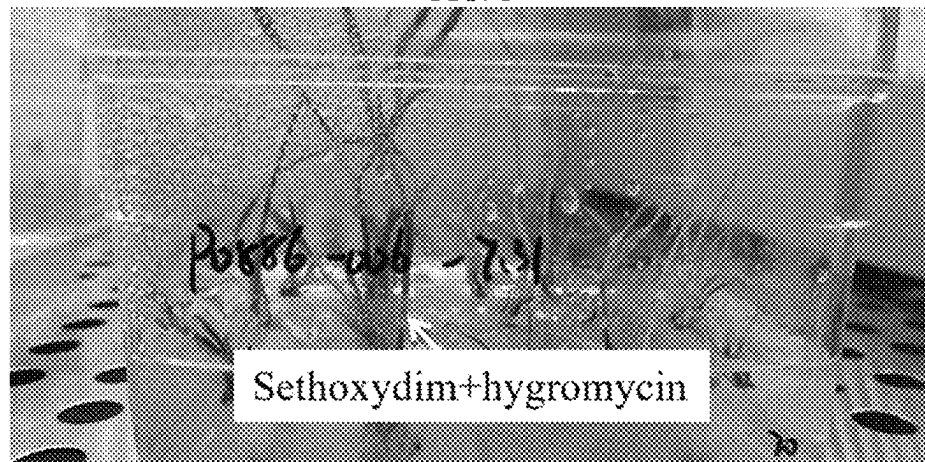


FIG. 2

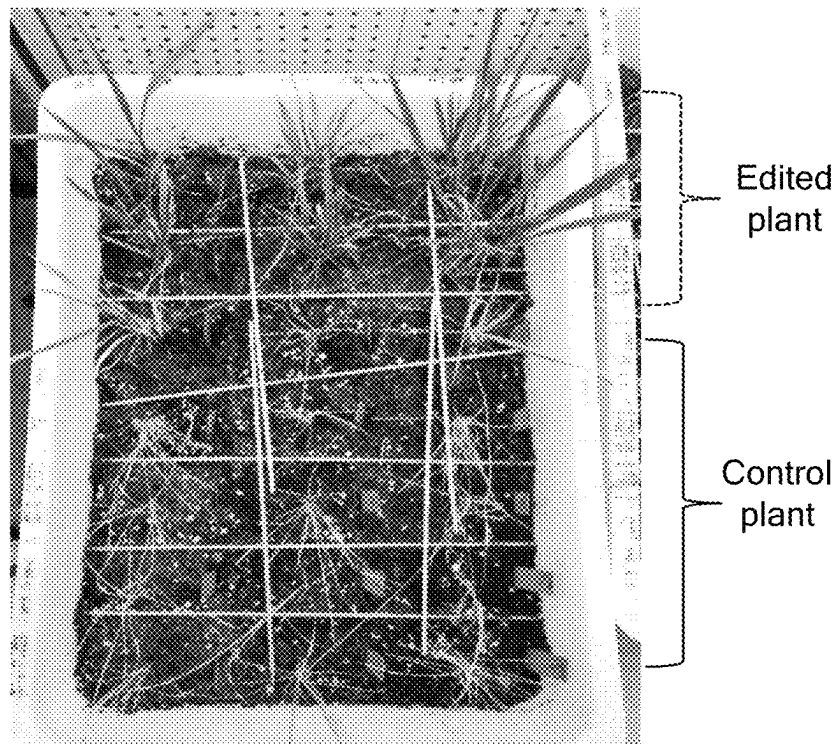


FIG. 3

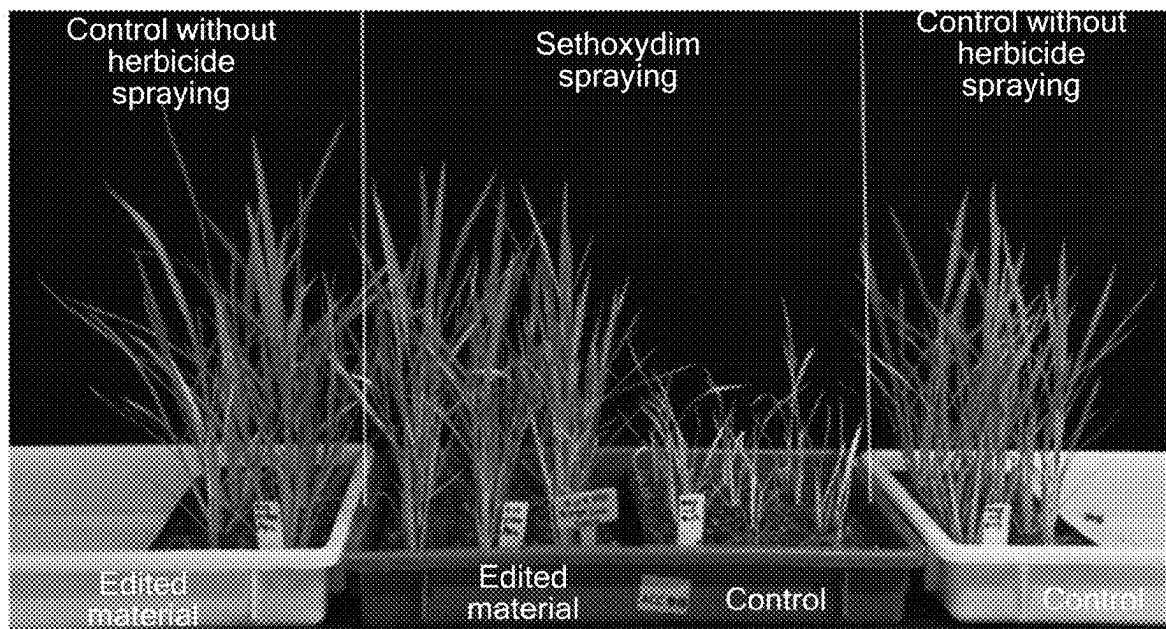


FIG. 4

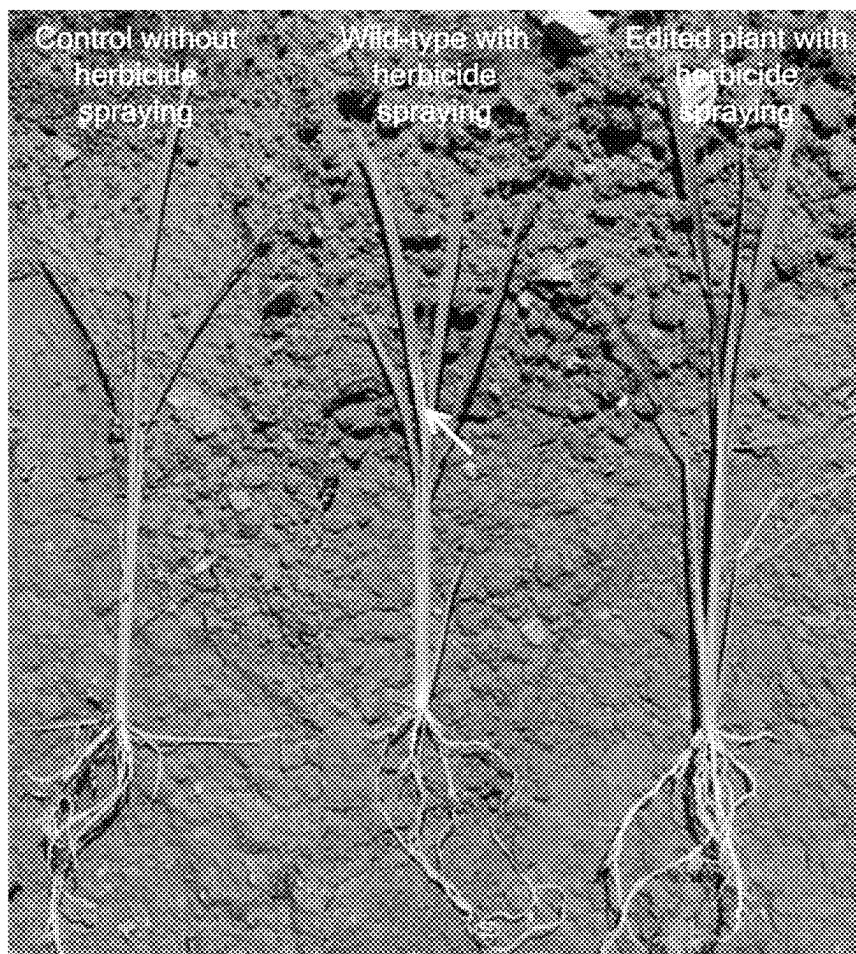
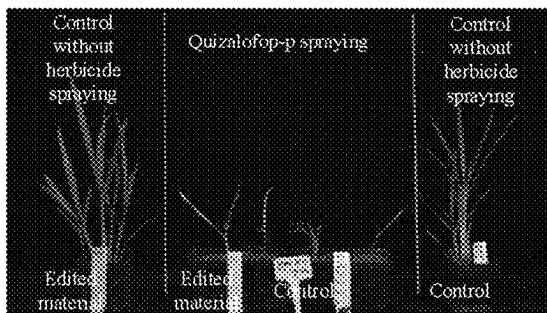
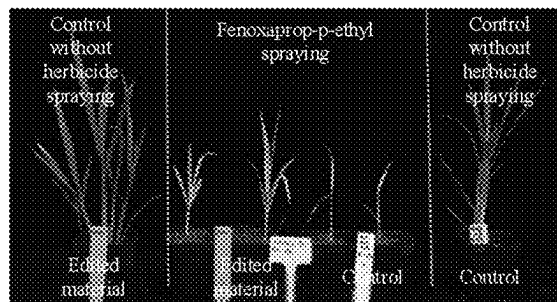


FIG. 5



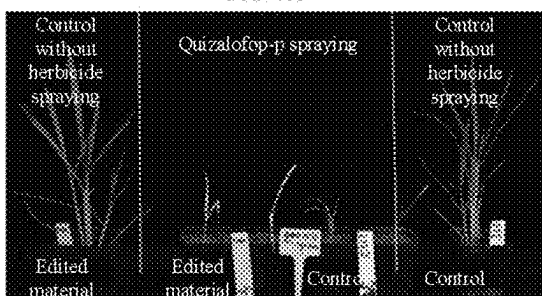
11879V-quizalofop-p

FIG. 6A



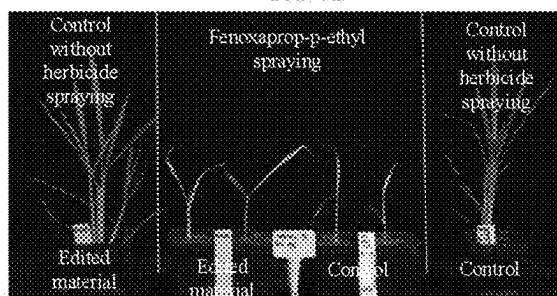
11879V-fenoxaprop-p-ethyl

FIG. 6B



C2186R-Quizalofop-p

FIG. 6C



C2186R-fenoxaprop-p-ethyl

FIG. 6D

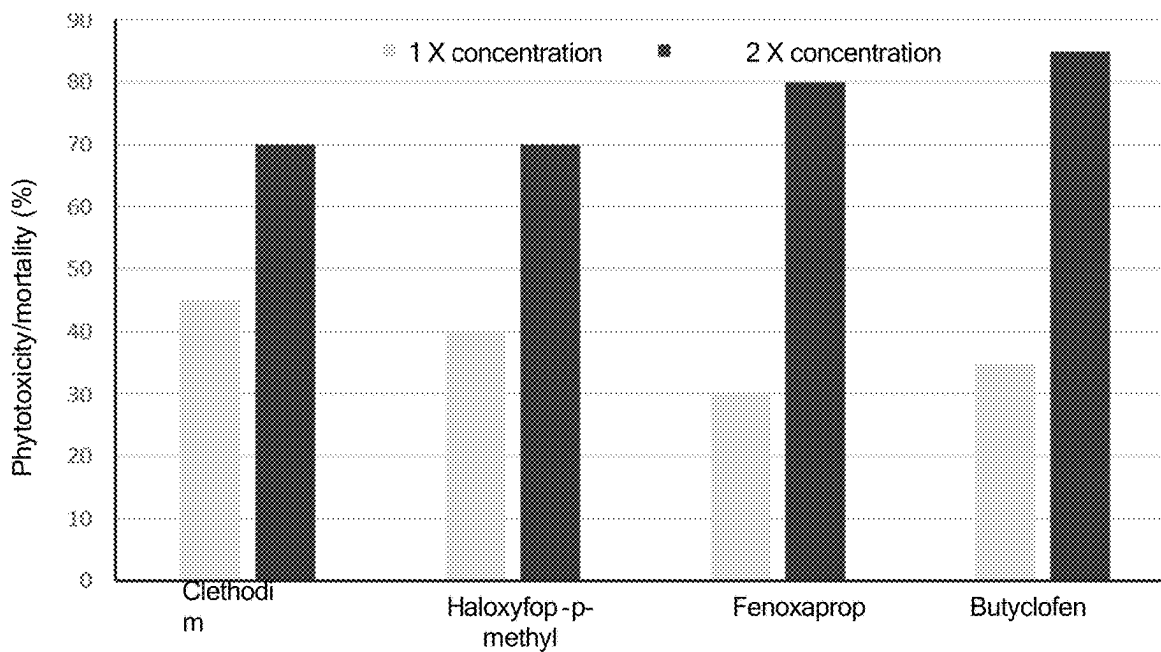


FIG. 7

1

HERBICIDE-RESISTANT ACETYL-CoA CARBOXYLASE (ACC) MUTANT AND USE THEREOF

TECHNICAL FIELD

The present disclosure belongs to the fields of biotechnology and crop genetic breeding, and specifically relates to an acetyl-CoA carboxylase (ACC) mutant protein, a nucleic acid encoding the ACC mutant protein, and a method and use for improving the resistance of a plant to a herbicide.

BACKGROUND

Oryza sativa is consumed by two-thirds of the world's population and is the main energy source in the diet of at least half of the two-thirds of the world's population. Rice is a low-cost food that is easily and rapidly prepared, which can be eaten together with various dishes.

The use of an herbicide to control weeds or plants in a crop has become an almost universal practice. As an important part of the modern agricultural production system, herbicides are considered as the most reliable and economical means among farmland weeding techniques. Since 2,4-dichlorophenoxyacetic acid (2,4-D) was first used in the 1940s, the herbicide industry has been developed for more than 60 years, and a large number of selective herbicides have been successfully developed. The research on acetyl-CoA carboxylase (ACC) inhibitors began in the 1970s. ACC herbicides are divided into the following 4 types: aryloxyphenoxypropionates (APPs), cyclohexanedione oximes (CHDs), aryloxyphenyl-cyclohexanediones (APCHDs), and cyclict rketones (CTRs). ACC herbicides can inhibit the synthesis of fatty acids in gramineous plants with high selectivity, can be transduced in plants, and can control annual or perennial gramineous weeds post-emergence. ACC herbicides have the advantages of high efficiency, low toxicity, long application period, safety to subsequent crops, and the like, and thus occupy an important position in the herbicide market.

ACC is an important target of chemical herbicides, which is a biotin-containing enzyme discovered in 1958. ACC catalyzes the carboxylation of acetyl-CoA in an organism to form malonyl-CoA, which provides a substrate for the synthesis of fatty acids and many secondary metabolites. ACC is a key enzyme or a rate-limiting enzyme for the biosynthesis of fatty acids. The carboxylase involves a two-step reversible reaction, including adenosine triphosphate (ATP)-dependent carboxylation of a biotin group on a substrate domain through biotin-carboxylase enzymatic activity, and transfer of carboxyl from biotin by carboxyl-transferase to obtain an acetyl-CoA substrate. ACC is a key enzyme for the biosynthesis of fatty acids in a plant, which occurs in chloroplasts and mitochondria. Moreover, ACC also plays a role in the formation of long-chain fatty acids and flavonoids and the malonylation in cytoplasm.

SUMMARY

The present disclosure is intended to provide a mutant ACC protein capable of imparting herbicide resistance to a plant or a polynucleotide encoding the same, and use thereof.

In the present disclosure, ACC refers to acetyl CoA carboxylase.

2

Mutant ACC

In an aspect, the present disclosure provides a mutant ACC, and compared to an amino acid sequence of a parent ACC, the mutant ACC has mutation at amino acid corresponding to amino acid 1,879 and/or amino acid 2,186 of an amino acid sequence shown in SEQ ID NO: 1.

In an embodiment, the amino acid 1,879 of the parent ACC may be isoleucine (I), and the amino acid 2,186 may be cysteine (C).

In an embodiment, the isoleucine (I) at position 1,879 may be mutated into an amino acid other than isoleucine (I), and the amino acid other than isoleucine (I) may be one or more selected from the group consisting of alanine (A), valine (V), glycine (G), leucine (L), glutamine (Q), phenylalanine (F), tryptophan (W), tyrosine (Y), aspartic acid (D), asparagine (N), glutamic acid (E), lysine (K), methionine (M), serine (S), threonine (T), cysteine (C), proline (P), histidine (H), and arginine (R).

In a preferred embodiment, the isoleucine (I) at position 1,879 may be mutated into valine (V).

In an embodiment, the cysteine (C) at position 2,186 may be mutated into an amino acid other than cysteine (C), and the amino acid other than cysteine (C) may be one or more selected from the group consisting of alanine (A), valine (V), glycine (G), leucine (L), isoleucine (I), phenylalanine (F), tryptophan (W), tyrosine (Y), aspartic acid (D), asparagine (N), lysine (K), glutamine (Q), methionine (M), serine (S), threonine (T), glutamic acid (E), proline (P), histidine (H), and arginine (R).

In a preferred embodiment, the cysteine (C) at position 2,186 may be mutated into arginine (R).

In an embodiment, the mutation may be selected from the group consisting of I1879V, C2186R, and a combination thereof.

In an embodiment, the parent ACC may be derived from any plant.

In an embodiment, the parent ACC may be derived from one or more selected from the group consisting of a gramineous plant, a leguminous plant, a chenopodiaceous plant, and a cruciferous plant.

In an embodiment, the parent ACC may be derived from one or more selected from the group consisting of *Arabidopsis thaliana* (*A. thaliana*), *Oryza sativa*, *Nicotiana tabacum*, *Zea mays*, *Sorghum bicolor*, *Hordeum vulgare*, *Triticum aestivum*, *Setaria italica*, *Glycine max*, *Lycopersicon esculentum*, *Solanum tuberosum*, *Chenopodium quinoa*, *Lactuca sativa*, *Brassica napus*, *Brassica pekinensis*, and *Fragaria ananassa*.

In a preferred embodiment, the parent ACC of the present disclosure may be derived from *Oryza* L., especially *Oryza sativa*.

In an embodiment, the parent ACC may have ACC activity, and an amino acid sequence of the parent ACC may have at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% sequence identity with the amino acid sequence shown in SEQ ID NO: 1.

In a preferred embodiment, the amino acid sequence of the parent ACC may have the sequence shown in SEQ ID NO: 1, or the amino acid sequence of the parent ACC may be shown in SEQ ID NO: 1.

In an embodiment, the mutant ACC may have at least 60%, preferably at least 70%, preferably at least 80%, and preferably at least 90% (such as 95%, 97%, or 99%) homology with a sequence shown in any one of SEQ ID NOS.: 2-4.

In an embodiment, the mutant ACC may be a polypeptide with an amino acid sequence shown in any one of SEQ ID NOs.: 2-4, an active fragment thereof, or a conservative variant polypeptide thereof.

In an embodiment, the mutant ACC may have an amino acid sequence shown in any one of SEQ ID NOs.: 2-4.

Fusion Protein

In another aspect, the present disclosure provides a fusion protein including the mutant ACC protein of the present disclosure. Further, the fusion protein may also include: a tag peptide, such as a histidine tag or 6×His; a plastid-targeted peptide, such as a chloroplast-targeted peptide; or a regulatory element, such as a promoter sequence, a terminator sequence, a leader sequence, a polyadenylation sequence, or a marker gene.

Polynucleotide

In another aspect, the present disclosure provides a polynucleotide encoding the mutant ACC protein or an active fragment thereof.

In an embodiment, the polynucleotide may be selected from the group consisting of:

(a) a polynucleotide encoding a protein shown in any one of SEQ ID NOs.: 2-4;

(b) a polynucleotide with a sequence shown in any one of SEQ ID NOs.: 5-7;

(c) a polynucleotide that has a nucleotide sequence of homology $\geq 80\%$ (preferably $\geq 90\%$, more preferably $\geq 95\%$, and most preferably $\geq 98\%$) with a sequence shown in SEQ ID NO: 5 and encodes a protein shown in SEQ ID NO: 2; or, a polynucleotide that has a nucleotide sequence of homology $\geq 80\%$ (preferably $\geq 90\%$, more preferably $\geq 95\%$, and most preferably $\geq 98\%$) with a sequence shown in SEQ ID NO: 6 and encodes a protein shown in SEQ ID NO: 3; or, a polynucleotide that has a nucleotide sequence of homology $\geq 80\%$ (preferably $\geq 90\%$, more preferably $\geq 95\%$, and most preferably $\geq 98\%$) with a sequence shown in SEQ ID NO: 7 and encodes a protein shown in SEQ ID NO: 4; and

(d) a polynucleotide complementary to any one selected from the group consisting of the polynucleotides described in (a) to (c).

In an embodiment, the polynucleotide may be selected from the group consisting of a genomic sequence, a cDNA sequence, an RNA sequence, and a combination thereof.

In an embodiment, the polynucleotide may preferably be single-stranded or double-stranded.

In an embodiment, the polynucleotide may make an auxiliary element selected from the group consisting of a signal peptide, a secretory peptide, a tag sequence (such as 6His), a nuclear localization signal (NLS), and a combination thereof additionally included at a flank of an open reading frame (ORF) of the mutant protein.

In an embodiment, the polynucleotide may further include a promoter operably linked to the ORF sequence of the mutant polypeptide.

In an embodiment, the promoter may be selected from the group consisting of a constitutive promoter, a tissue-specific promoter, an inducible promoter, and a strong promoter.

Nucleic Acid Construct

In another aspect, the present disclosure provides a nucleic acid construct, including the polynucleotide and a regulatory element operably linked thereto.

In an embodiment, the regulatory element may be one or more selected from the group consisting of an enhancer, a transposon, a promoter, a terminator, a leader sequence, a polyadenylation sequence, and a marker gene.

Vector

The present disclosure also provides a vector including a nucleic acid sequence encoding the mutant ACC or the fusion protein of the present disclosure. Preferably, the vector may further include an expression regulation element operably linked to the aforementioned nucleic acid sequence.

In an embodiment, the vector may include a cloning vector, an expression vector, a shuttle vector, and an integration vector.

In an embodiment, the vector may be a vector for gene editing of the endogenous ACC gene in a host cell.

In an embodiment, the vector may include a polynucleotide encoding a polypeptide shown in any one of SEQ ID NOs.: 2-4.

In an embodiment, the expression vector may also include at least one replication origin to realize self-replication.

In an embodiment, the vector may be a vector that will be integrated into a genome of a host cell when introduced into the host cell and then replicates together with a chromosome into which the vector is integrated.

The vector can be, for example, a plasmid, a virus, a cosmid, a phage, and the like, which are well known to those skilled in the art.

Preferably, the vector in the present disclosure may be a plasmid.

Edit Vector System

In another aspect, the present disclosure provides an edit vector system that can produce the above-mentioned mutant ACC in a plant. The edit vector system may include one or more vectors, and the one or more vectors may at least include a guide sequence targeting the parent ACC.

The guide sequence may include a part of the nucleotide sequence of the parent ACC, preferably at least 15 bp of the ACC nucleotide sequence and more preferably at least 20 bp of the ACC nucleotide sequence. In an embodiment, the edit vector may further include a gene editing enzyme. The gene editing enzyme may include nucleases suitable for CRISPR, transcription activator-like (TAL) effector nuclease (TALEN), zinc-finger nuclease (ZFN), and other editing tools.

Preferably, the gene editing enzyme may be a Cas protein, also known as a CRISPR enzyme or a Cas effector protein, including but not limited to: a Cas9 protein, a Cas12 protein, a Cas13 protein, a Cas14 protein, a Csm1 protein, and an FDK1 protein.

Preferably, the Cas protein may be operably linked to a first regulatory element.

In an embodiment, the gene editing enzyme may be a Cas9 protein, and the vector may further include a Scaffold sequence that can specifically bind to the Cas9 protein. The Scaffold sequence and the guide sequence are operably linked to form a gRNA. Preferably, the gRNA may be operably linked to a second regulatory element.

In other embodiments, the gene editing enzyme may be a Cas12 protein, such as Cas12a, Cas12b, and Cas12i, and the vector may further include a direct repeat that specifically binds to the Cas12 protein. The direct repeat and the guide sequence are operably linked to form a gRNA. Preferably, the gRNA may be operably linked to the second regulatory element.

The above-mentioned regulatory elements may include a promoter, a terminator sequence, a leader sequence, a polyadenylation sequence, a signal peptide coding region, a marker gene, an enhancer, an internal ribosome entry site (IRES), and other expression control elements (for example, a transcriptional termination signal, such as a polyadenylation signal and a poly-U sequence).

Preferably, the edit vector system may further include a base editing element, which is adenine deaminase and/or cytosine deaminase.

In an embodiment, the edit vector may also include a resistance gene to facilitate screening; and the resistance gene may include hyg, bar, kana, rif, spec, and amp, which are well known to those skilled in the art.

Preferably, the guide sequence of gRNA may be A-ACC1879: GAGAATATACATGGAAGTGC; A-ACC2186: ATAGCACTCAATGCGGTCTG; or a combination thereof.

Preferably, the Cas protein may be nCas9 or another Cas9 protein with nick activity. The “n” stands for nick, that is, the “n” indicates a Cas protein with only single-strand cleavage activity.

Host Cell

In another aspect, the present disclosure provides a host cell, where the host cell includes one or more selected from the group consisting of the mutant ACC, the gene encoding the mutant ACC, the fusion protein, the vector, and the nucleic acid construct; or, the polynucleotide is integrated into a genome of the host cell.

In an embodiment, the host cell may be a prokaryotic cell, such as *Escherichia coli* (*E. coli*).

In an embodiment, the host cell may be a plant cell, and the plant may include an angiosperm and a gymnosperm.

In an embodiment, the plant may include a monocotyledonous plant and a dicotyledonous plant.

In an embodiment, the plant may include an herbaceous plant and a woody plant.

In an embodiment, the plant may include *A. thaliana*, *Nicotiana tabacum*, *Oryza sativa*, *Zea mays*, *Sorghum bicolor*, *Hordeum vulgare*, *Triticum aestivum*, *Setaria italica*, *Glycine max*, *Lycopersicon esculentum*, *Solanum tuberosum*, *Chenopodium quinoa*, *Lactuca sativa*, *Brassica napus*, *Brassica pekinensis*, and *Fragaria ananassa*.

Resistant Plant

In another aspect, the present disclosure provides an herbicide-resistant plant, where the herbicide-resistant plant includes one or more selected from the group consisting of the mutant ACC, the polynucleotide encoding the mutant ACC, the fusion protein, the vector, and the nucleic acid construct; or, the polynucleotide is integrated into a genome of the plant.

The term “substantially identical” means that two amino acid sequences have 85%, 90%, 95%, 96%, 97%, 98%, or 99% identity.

Method for Preparing the Mutant Polypeptide

In another aspect, the present disclosure provides a method for preparing the mutant ACC polypeptide or an active fragment thereof, including the following step:

(a) under conditions suitable for expression, cultivating a host cell including the mutant ACC polypeptide to express the mutant ACC polypeptide; and preferably, the method may further include:

(b) isolating the mutant ACC polypeptide.

Method for Acquiring the Herbicide-Resistant Plant

In another aspect, the present disclosure provides a plant cell, a plant seed, a plant tissue, a plant part, or a plant with herbicide resistance, where the plant cell, the plant tissue, the plant seed, the plant part, or the plant includes the mutant ACC polypeptide or a polynucleotide sequence encoding the same.

In another aspect, the present disclosure provides a method for acquiring or preparing a plant cell, a plant seed, a plant tissue, a plant part, or a plant with herbicide resistance, including: introducing the mutant ACC polypeptide or

a polynucleotide sequence encoding the same into the plant cell, the plant seed, the plant tissue, the plant part, or the plant.

In an embodiment, the introducing the ACC mutant polypeptide of the present disclosure may include the following step: allowing the ACC mutant polypeptide to express in the plant cell, the plant seed, the plant tissue, the plant part, or the plant. For example, the mutant polypeptide is expressed by an expression vector, or the mutant polypeptide is integrated into a plant genome for expression.

In a preferred embodiment, the above method may include the following steps:

(1) providing *Agrobacterium tumefaciens* (*A. tumefaciens*) carrying an expression vector, where the expression vector includes a DNA coding sequence of the mutant ACC polypeptide or an active fragment thereof;

(2) contacting the plant cell, the plant tissue, or the plant part with the *A. tumefaciens* in step (1), such that the DNA coding sequence of the mutant ACC polypeptide or the active fragment thereof is transformed into the plant cell and integrated on a chromosome of the plant cell; and

(3) screening out a plant cell transformed with the DNA coding sequence of the mutant ACC polypeptide or the active fragment thereof.

In an embodiment, the introducing the mutant ACC polypeptide may include the following step: allowing endogenous ACC of the plant to mutate to introduce the mutant polypeptide. Preferably, the mutant polypeptide may be introduced by gene editing.

In another preferred embodiment, the method may include: allowing an endogenous ACC coding sequence in the plant cell, the plant seed, the plant tissue, or the plant part to mutate at positions corresponding to amino acid 1,879 and/or amino acid 2,186 of SEQ ID NO: 1.

In another preferred embodiment, the method may include following steps:

(1) introducing the aforementioned edit vector system into the plant cell, the plant seed, the plant tissue, or the plant part; and

(2) allowing a gene editing tool to act on an endogenous ACC coding sequence, such that the endogenous ACC coding sequence mutates at positions corresponding to amino acid 1,879 and/or amino acid 2,186 of the amino acid sequence shown in SEQ ID NO: 1.

Further, the above method may also include the following step: screening a mutant plant cell, plant tissue, or plant part, and optionally, isolating the gene editing tool.

In another preferred embodiment, the gene editing tool may include CRISPR, TALEN, and ZFN.

In another preferred embodiment, the plant may include an angiosperm and a gymnosperm.

In another preferred embodiment, the plant may include a monocotyledonous plant and a dicotyledonous plant.

In another preferred embodiment, the plant may include an herbaceous plant and a woody plant.

In another preferred embodiment, the plant may include *A. thaliana*, *Nicotiana tabacum*, *Oryza sativa*, *Zea mays*, *Sorghum bicolor*, *Hordeum vulgare*, *Triticum aestivum*, *Setaria italica*, *Glycine max*, *Lycopersicon esculentum*, *Solanum tuberosum*, *Chenopodium quinoa*, *Lactuca sativa*, *Brassica napus*, *Brassica pekinensis*, and *Fragaria ananassa*.

In an embodiment, compared with the maximum tolerated herbicide concentration of the parent plant, the maximum tolerated herbicide concentration of the plant with the mutant polypeptide may be increased by at least 1 to 4 times, such as 2, 3, or 4 times. When an herbicide is applied, the

plant with the mutant polypeptide of the present disclosure can tolerate the herbicide at a concentration least 2 to 4 times a recommended concentration of the herbicide, while the parent plant cannot tolerate the herbicide at this concentration.

Method for Controlling Weeds

In another aspect, the present disclosure also provides a method for controlling the growth of weeds near a plant, including:

a) providing the above-mentioned herbicide-resistant plant; and

b) applying an effective amount of the herbicide to the plant and weeds nearby to control the weeds.

The plant may preferably be *Oryza sativa*.

The herbicide may include one or more selected from the group consisting of APP, CHD, APCHD, and CTR. Preferably, the herbicide may include one or more selected from the group consisting of alloxydim, butoxydim, clethodim, cyclohexenone, cycloxydim, sethoxydim, tepraloxym, tralkoxydim, phentodim, chloroazifop-propynyl, clodinafop-propargyl, clofop, diclofop-methyl, fenoxaprop-ethyl, fenoxaprop-p-ethyl, fenthiaprop-ethyl, fluazifop-butyl, fluazifop-p-butyl, haloxyfop-methyl, haloxyfop-p-methyl, clomazone, clefoxidim, propaquizafop, quizalofop, quizalofop-p, quizalofop-p-ethyl, quizalofop-p-tefuryl, trifluorophenoxy propionic acid, fenoxaprop, and a salt or ester thereof.

Preferably, the herbicide may be one or more selected from the group consisting of sethoxydim, phentodim, and clethodim.

Use

In another aspect, the present disclosure provides use of the mutant ACC, the gene encoding the mutant ACC, the fusion protein, the vector, or the nucleic acid construct in a reagent or a kit for preparing an herbicide-resistant plant.

In another aspect, the present disclosure provides use of the mutant ACC, the gene encoding the mutant ACC, the fusion protein, the vector, or the nucleic acid construct for controlling weeds.

In another aspect, the present disclosure provides use of the mutant ACC, the gene encoding the mutant ACC, the fusion protein, the vector, or the nucleic acid construct in the preparation of an herbicide-resistant plant.

Herbicide

In an embodiment, the herbicide of the present disclosure may be an ACC-inhibiting herbicide, and the ACC-inhibiting herbicide includes, but is not limited to, one or more selected from the group consisting of APP, CHD, APCHD, and CTR.

Preferably, the herbicide of the present disclosure includes, but is not limited to, one or more selected from the group consisting of alloxydim, butoxydim, clethodim, cyclohexenone, cycloxydim, sethoxydim, tepraloxym, tralkoxydim, phentodim, chloroazifop-propynyl, clodinafop-propargyl, clofop, diclofop-methyl, fenoxaprop-ethyl, fenoxaprop-p-ethyl, fenthiaprop-ethyl, fluazifop-butyl, fluazifop-p-butyl, haloxyfop-methyl, haloxyfop-p-methyl, clomazone, clefoxidim, propaquizafop, quizalofop, quizalofop-p, quizalofop-p-ethyl, quizalofop-p-tefuryl, trifluorophenoxy propionic acid, fenoxaprop, and a salt or ester thereof. Preferably, the herbicide may be one or more selected from the group consisting of phentodim, sethoxydim, and clethodim.

General Definitions

Unless defined otherwise, the technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art.

The terms “polynucleotide”, “nucleotide sequence”, “nucleic acid sequence”, “nucleic acid molecule”, and “nucleic acid” may be used interchangeably and include DNA, RNA, or a hybrid thereof, which may be double-stranded or single-stranded.

The term “homology” or “identity” used refers to sequence matching between two polypeptides or between two nucleic acids. Therefore, the composition and method of the present disclosure also include homologues of the nucleotide sequence and the polypeptide sequence (such as SEQ ID NOs: 1-7) of the present disclosure. “Homology” can be calculated by a known method including but not limited to: Computational Molecular Biology (edited by Lesk, A. M.), Oxford University Press, New York (1988); Biocomputing: Informatics and Genome Projects (edited by Smith, D. W.), Academic Press, New York (1993); Computer Analysis of Sequence Data, Part I (edited by Griffin, A. M. and Griffin, H. G.), Humana Press, New Jersey (1994); Sequence Analysis in Molecular Biology (edited by von Heinje, G.), Academic Press (1987); and Sequence Analysis Primer (edited by Gribbskov, M. and Devereux, J.), Stockton Press, New York (1991).

A specific amino acid position (number) in the protein of the present disclosure is determined by aligning an amino acid sequence of the target protein with SEQ ID NO: 1 using a standard sequence alignment tool. For example, the Smith-Waterman algorithm or CLUSTALW2 algorithm is used to align two sequences, and the sequences are considered aligned when an alignment score is the highest. The alignment score can be calculated according to the method described in Wilbur, W. J. and Lipman, D. J. (1983) Rapid similarity searches of nucleic acid and protein data banks. Proc. Natl. Acad. Sci. USA, 80: 726-730. In the ClustalW2 (1.82) algorithm, default parameters may be preferably used: protein gap opening penalty=10.0; protein gap extension penalty=0.2; protein matrix=Gonnet; protein/DNA end gap=-1; and protein/DNAGAPDIST=4. The AlignX program (a part of the vectorNTI group) may preferably be used to align an amino acid sequence of a protein with SEQ ID No: 1 using default parameters suitable for multiple alignments (gap opening penalty: 10 og; and gap extension penalty: 0.05) to determine a position of a specific amino acid within the protein.

The term “encoding” refers to an inherent characteristic of a specific nucleotide sequence in a polynucleotide, such as a gene cDNA or mRNA, which serves as a template for the synthesis of a defined nucleotide sequence (namely, rRNA, tRNA, and mRNA) or a defined amino acid sequence and the synthesis of other polymers and macromolecules in a biological process of a biological characteristic thereof. Therefore, if the transcription and translation of mRNA corresponding to a gene produces a protein in a cell or another biological system, the gene encodes the protein.

The term “amino acid” refers to a carboxylic acid with amino. Various proteins in organisms are composed of 20 essential amino acids.

The terms “protein”, “polypeptide”, and “peptide” can be used interchangeably in the present disclosure and refer to a polymer of amino acid residues, including a polymer in which one or more amino acid residues are a chemical analogue of a natural amino acid residue. The protein and polypeptide of the present disclosure can be produced through recombination or chemical synthesis.

The term “mutant protein” refers to a protein that is obtained through substitution, insertion, deletion, and/or addition of one or more amino acid residues based on an amino acid sequence of a parent protein.

The term "AxxB" means that amino acid A at position xx is changed into amino acid B. For example, I1879V means that I at position 1,879 is changed into V, C2186R means that C at position 2,186 is changed into R, and so on. For double or multiple mutations, mutations are separated by a "/". For example, I1879V/C2186R indicates that, relative to the amino acid sequence of SEQ ID NO: 1, I at position 1,879 is substituted by V and C at position 2,186 is substituted by R, where both mutations are present in the specific mutant ACC protein.

As used herein, the term "regulatory element" is intended to include a promoter, a terminator sequence, a leader sequence, a polyadenylation sequence, a signal peptide coding region, a marker gene, an enhancer, an IRES, and other expression control elements (for example, a transcriptional termination signal, such as a polyadenylation signal and a poly-U sequence), and the detailed description can be seen in Goeddel, "GENE EXPRESSION TECHNOLOGY: METHODS IN ENZYMOLOGY" 185, Academic Press, San Diego, Calif. (1990). In some cases, the regulatory element includes sequences that guide the constitutive expression of a nucleotide sequence in many types of host cells and sequences that guide the expression of the nucleotide sequence only in some host cells (such as a tissue-specific regulatory sequence). A tissue-specific promoter can mainly guide the expression in a desired tissue of interest, such as muscles, neurons, bones, skin, blood, specific organs (such as liver and pancreas), or specific cell types (such as lymphocytes). In some cases, a regulatory element can also guide the expression in a time-dependent manner (such as in a cell cycle-dependent or developmental stage-dependent manner), which may be or may not be tissue or cell type-specific. In some cases, the term "regulatory element" covers enhancer elements, such as WPRE; CMV enhancer; R-U5' fragment in LTR of HTLV-I ((Mol. Cell. Biol., Vol 8 (1): 466-472, 1988); SV40 enhancer; and an intron sequence between exons 2 and 3 of rabbit β -globin (Proc. Natl. Acad. Sci. USA., Vol. 78 (3): 1527-31, 1981).

As used herein, the term "promoter" has the meaning well known to those skilled in the art, which refers to a non-coding nucleotide sequence located upstream of a gene and capable of promoting the expression of a downstream gene. A constitutive promoter is a nucleotide sequence that will result in the generation of a gene product in a cell under most or all physiological conditions of the cell after the promoter is operably linked to a polynucleotide encoding or defining the gene product. An inducible promoter is a nucleotide sequence that will cause the generation of a gene product in a cell only when there is an inducer corresponding to the promoter in the cell after the promoter is operably linked to a polynucleotide encoding or defining the gene product. A tissue-specific promoter is a nucleotide sequence that will cause the generation of a gene product in a cell basically only when the cell is a cell of the tissue type corresponding to the promoter after the promoter is operably linked to a polynucleotide encoding or defining a gene product.

"NLS" (Nuclear Localization Signal) is an amino acid sequence that tags a protein for import into the nucleus through nuclear transport, that is, a protein with NLS is transported to the nucleus. Typically, NLS may include positively charged Lys or Arg residues that are exposed on the surface of a protein. Exemplary NLS includes, but is not limited to, SV40 large T antigen, EGL-13, c-Myc, and TUS protein.

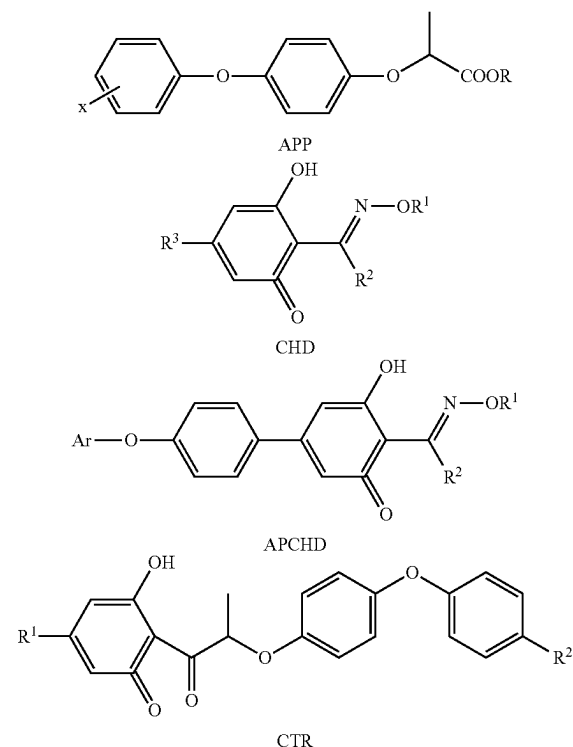
As used herein, the term "operably linked" means that a nucleotide sequence of interest is linked to one or more regulatory elements in a manner that allows the expression

of the nucleotide sequence (for example, in an in vitro transcription/translation system or in a host cell when the vector is introduced into the host cell).

The term "vector" refers to an element that is allowed to be integrated into a genome of a host cell or to self-replicate within a cell independently of its genome. The vector may include any elements that guarantee the self-replication. The vector usually carries a gene that is not a part of the central metabolism of a cell and is usually in the form of double-stranded DNA. The selection of a vector generally depends on the compatibility of the vector with a host cell into which the vector is to be introduced. When a vector needs to be used, the selection of the vector depends on a method for transforming a host cell well known to those skilled in the art. For example, a plasmid vector can be used.

The term "ACC herbicide" refers to an herbicide that can inhibit the synthesis of fatty acids in gramineous plants with high selectivity, can be transduced in plants, and can control annual or perennial gramineous weeds post-emergence. The ACC herbicides have the advantages of high efficiency, low toxicity, long application period, safety to subsequent crops, and the like, and thus occupy an important position among herbicides.

The ACC herbicides can be divided into the following 4 types: APPs, CHDs, APCHDs, and CTRs, which are shown as follows.



The herbicide includes, but is not limited to, alloxydim, butoxydim, clethodim, cyclohexenone, cycloxydim, sethoxydim, tepraloxym, tralkoxydim, chloroazifop-propynyl, clodinafop-propargyl, clofop, diclofop-methyl, fenoxaprop-ethyl, fenoxaprop-p-ethyl, fenthiaprop-ethyl, fluazifop-butyl, fluazifop-p-butyl, haloxyfop-methyl, haloxyfop-p-methyl, clomazone, clefoxidim, propaquizafop, quizalofop, quizalofop-p, quizalofop-p-ethyl, quizalofop-p-tefuryl, trifluorophenoxy propionic acid, and fenoxaprop.

The “herbicide resistance” refers to the genetic ability of a plant to survive and reproduce after the plant is exposed to an herbicide at a dosage that is usually lethal to the wild-type plant. In plants, the herbicide resistance can be naturally produced or induced by a technique such as genetic engineering or screening a variant produced by tissue culture or mutagenesis. Unless otherwise stated, the herbicide resistance is heritable and allows a plant to grow and reproduce in the presence of a typical effective herbicidal treatment for a given plant by the herbicide, as suggested by the current version of “Herbicide Manual” when the present disclosure is filed. As recognized by those skilled in the art, a plant can still be considered “resistant”, even if a specified degree of plant damage due to herbicide exposure is obvious. As used herein, the term “tolerance” includes “resistance” or “resistant plant” as defined herein, and the improved ability of a specific plant to tolerate various degrees of damage induced by an herbicide (typically ethyl) at the same herbicide dosage as the wild-type plant of the same genotype.

The term “parent ACC polypeptide” refers to a polypeptide derived from the ACC mutant polypeptide. In a preferred embodiment, the parent ACC polypeptide is a protein (polypeptide) that can be found in nature or is encoded by a nucleic acid that can be found in nature, where nucleotides of the nucleic acid can be obtained through genetic engineering such as genome sequencing and polymerase chain reaction (PCR), and an amino acid sequence of the protein can be deduced from the nucleotide sequence. An amino acid sequence of the wild-type ACC polypeptide is shown in SEQ ID NO: 1, for example. In some embodiments, the parent ACC polypeptide may be obtained by changing one or more amino acid residues of the wild-type ACC polypeptide without affecting the enzymatic activity.

The terms “mutant ACC protein”, “mutant ACC”, “mutant ACC enzyme”, “mutant protein”, “mutant polypeptide”, “polypeptide of the present disclosure”, “protein of the present disclosure”, and the like can be used interchangeably. Preferably, the mutant protein may be obtained through mutation of amino acid 1,879 and/or amino acid 2,186 of the sequence shown in SEQ ID NO: 1.

The term “host organism” should be understood as any unicellular or multicellular organism into which a nucleic acid encoding the mutant ACC protein can be introduced, including, for example, bacteria such as *E. coli*, fungi such as yeast (such as *Saccharomyces cerevisiae* (*S. cerevisiae*)), molds (such as *Aspergillus*), plant cells, and plants.

The term “plant” should be understood as any differentiated multicellular organism capable of photosynthesis, including: crop plants at a mature or developmental stage, especially monocotyledonous or dicotyledonous plants; vegetable crops including artichoke, turnip cabbage, arugula, leek, asparagus, lettuce (such as head lettuce, leaf lettuce, and romaine lettuce), bok choy, malanga, melons (such as cantaloupe, watermelon, crenshaw melon, honeydew melon, and Roman cantaloupe), rape crops (such as Brussels sprout, cabbage, cauliflower, broccoli, borecole, kale, Chinese cabbage, and bok choy), cardoon, carrot, napa, okra, onion, celery, parsley, chickpea, parsnip, chicory, pepper, *Solanum tuberosum*, gourd (such as marrow squash, cucumber, zucchini, cushaw, and pumpkin), radish, dried ball onion, rutabaga, purple eggplant (also known as eggplant), salsify, lettuce, shallot, endive, garlic, spinach, green onion, cushaw, greens, beets (sugar beets and fodder beets), sweet potato, Swiss chard, wasabi, tomato, turnip, and spices; fruits and/or vine crops such as apple, apricot, cherry, nectarine, peach, pear, plum, prune, cherry, quince, almond, chestnut, hazelnut, pecan, pistachio, walnut, citrus, blueberry, boysenberry,

cranberry, currant, loganberry, raspberry, strawberry, blackberry, grape, avocado, banana, kiwi, persimmon, pomegranate, pineapple, tropical fruit, pome, melon, mango, papaya, and lychee; field crops, such as clover, alfalfa, evening primrose, meadowfoam, corn/maize (forage corn, sweet corn, and popcorn), lupulus, jojoba, peanut, rice, safflower, small grain crops (*Hordeum vulgare*, oat, rye, *Triticum aestivum*, and the like), *Sorghum bicolor*, *Nicotiana tabacum*, kapok, legumes (beans, lentil, pea, and *Glycine max*), oil plants (canola, leaf mustard, poppy, olive, sunflower, coconut, castor oil plant, cocoa bean, and groundnut), *Ara-bidopsis*, fiber plants (cotton, flax, hemp, and jute), *Lauraceae* (cinnamon or camphor), or a plant such as coffee, sugar cane, tea, and natural rubber plants; and/or bedding plants such as a flowering plant, cactus, a succulent plant, and/or an ornamental plant, and trees such as forests (broad-leaved and evergreen trees, such as conifers), fruit trees, ornamental trees, nut-bearing trees, shrubs, and other seedlings.

The term “plant tissue” or “plant part” includes a plant cell, a protoplast, a plant tissue culture, a plant callus, a plant piece, a plant embryo, a pollen, an ovule, a seed, a leaf, a stem, a flower, a branch, a seedling, a fruit, a nucleus, a spike, a root, a root tip, an anther, and the like.

The term “plant cell” should be understood as any cell derived or found in a plant, which is capable of forming, for example, undifferentiated tissues such as calli, differentiated tissues such as embryos, constituent parts of a plant, plants, or seeds.

The gene editing technology includes CRISPR technology, TALEN technology, and ZFN technology. “CRISPR” refers to clustered regularly interspaced short palindromic repeat, which comes from the immune system of microorganisms. A gene editing tool includes gRNA and Cas protein (such as Cas9, Cpf1, and Cas12b). The gene editing tool of TALEN refers to a restriction enzyme that can cleave a specific DNA sequence, which includes a TAL effector DNA binding domain and a DNA cleavage domain. The gene editing tool of ZFN refers to a restriction enzyme that can cleave a specific DNA sequence, which includes a zinc-finger DNA binding domain and a DNA cleavage domain. It is well known to those skilled in the art that an intracellular genome can be edited by constructing nucleotides encoding a gene editing tool and other regulatory elements into a suitable vector and then transforming the vector into a cell; and a type of the editing includes gene knockout, insertion, and base editing.

As used herein, the term “gene editing enzyme” refers to a nuclease suitable for CRISPR, TALEN, ZFN, and other editing tools. Preferably, the gene editing enzyme may be a CRISPR enzyme, also known as a Cas protein, including but not limited to: a Cas9 protein, a Cas12 protein, a Cas13 protein, a Cas14 protein, a Csm1 protein, and an FDK1 protein. The Cas protein refers to a protein family. Cas proteins have different structures according to different sources, such as SpCas9 derived from *Streptococcus pyogenes* (*S. pyogenes*) and SaCas9 derived from *Staphylococcus aureus* (*S. aureus*). Cas proteins can also be classified based on structural characteristics (such as domains), for example, the Cas12 family may include Cas12a (also known as Cpf1), Cas12b, Cas12c, Cas12i, and the like. The Cas protein may be double-stranded or single-stranded or may have no cleavage activity. The Cas protein of the present disclosure may be wild-type or a mutant thereof. A mutation type of the mutant may include amino acid substitution, substitution, or deletion. The mutant may or may not change the enzyme cleavage activity of the Cas protein. Preferably,

13

the Cas protein of the present disclosure may only have single-strand cleavage activity or may not have cleavage activity, which may be a mutant of the wild-type Cas protein. Preferably, the Cas protein of the present disclosure may be Cas9, Cas12, Cas13, or Cas14 with single-strand cleavage activity. In a preferred embodiment, the Cas9 protein of the present disclosure may include SpCas9n(D10A), nSpCas9NG, SaCas9n, ScCas9n, and XCas9n, where "n" stands for nick, indicating a Cas protein with only single-strand cleavage activity. The mutation of a known Cas protein to obtain a Cas protein with single-strand cleavage activity or without cleavage activity is a conventional technique in the art. Those skilled in the art know that a variety of Cas proteins with nucleic acid cleavage activity have been reported in the art; and the known proteins or modified variants can all achieve the functions of the present disclosure, which are incorporated into the protection scope of the present disclosure by reference.

For example, it is clear to those skilled in the art that a structure of a protein can be changed without adversely affecting the activity and functionality of the protein. For example, one or more conservative amino acid substitutions can be introduced into an amino acid sequence of a protein without adversely affecting the activity and/or three-dimensional (3D) configuration of the protein molecule. Those skilled in the art are aware of examples and implementations of the conservative amino acid substitutions. Specifically, an amino acid residue can be substituted by another amino acid residue that belongs to the same group as the amino acid residue to be substituted. That is, a nonpolar amino acid residue can be substituted by another nonpolar amino acid residue; an uncharged polar amino acid residue can be substituted by another uncharged polar amino acid residue; a basic amino acid residue can be substituted by another basic amino acid residue; and an acidic amino acid residue can be substituted by another acidic amino acid residue. Such substituted amino acid residues may be or may not be encoded by genetic codes. As long as a substitution does not damage the biological activity of a protein, a conservative substitution in which an amino acid is substituted by another amino acid belonging to the same group falls within the scope of the present disclosure. Therefore, in addition to the above-mentioned mutations, the mutant GBSS1 protein of the present disclosure may include one or more other mutations such as conservative substitutions in the amino acid sequence. In addition, the present disclosure covers mutant ACC proteins with one or more other non-conservative substitutions, as long as the non-conservative substitutions do not significantly affect the desired function and biological activity of the protein of the present disclosure.

As well known in the art, one or more amino acid residues can be deleted from the N and/or C terminus of a protein while still retaining its functional activity. Therefore, in another aspect, the present disclosure also relates to a fragment that is obtained through deletion of one or more amino acid residues from the N-terminus and/or C-terminus of the mutant ACC protein and retains the required functional activity (such as an amino acid fragment with the mutation site of the present disclosure), which is also within the scope of the present disclosure and is called a biologically-active fragment. In the present disclosure, the "biologically-active fragment" refers to a part of the mutant ACC protein of the present disclosure, which retains the biological activity of the mutant ACC protein of the present disclosure. For example, the biologically-active fragment of the mutant ACC protein may be obtained after one or more (for example, 1-50, 1-25, 1-10, or 1-5, such as 1, 2, 3, 4, or

14

5) amino acid residues are deleted from the N-terminus and/or C-terminus of the protein, which still retains the biological activity of the full-length protein.

In addition, the mutant protein of the present disclosure can be modified. Modified (usually without changing the primary structure) forms may include: chemically derived forms of the mutant protein in vivo or in vitro, such as acetylated or carboxylated form. The modification may also include glycosylation, such as glycosylation modification during the synthesis and processing or further processing of the mutant protein to produce a mutant protein. The modification can be accomplished by exposing the mutant protein to a glycosylase (such as a mammalian glycosylase or deglycosylase). The modified forms may also include sequences with phosphorylated amino acid residues (such as phosphotyrosine, phosphoserine, and phosphothreonine). The mutant protein can also be modified to increase its proteolysis resistance or optimize its solubility.

The present disclosure also provides a polynucleotide encoding the mutant ACC polypeptide, which may also include additional coding and/or non-coding sequences. Preferably, the mutant ACC polypeptide may be shown in SEQ ID NOs: 2-4. Those skilled in the art are well aware that, due to the degeneracy of genetic codes, there are many different nucleic acid sequences that can encode the amino acid sequence disclosed herein. Producing other nucleic acid sequences encoding the same protein is within the competence scope of those of ordinary skill in the art, and thus the present disclosure covers nucleic acid sequences encoding the same amino acid sequence due to the degeneracy of genetic codes. For example, in order to achieve the high expression of a heterologous gene in a target host organism such as a plant, the gene can be optimized using a codon preferred by the host organism to allow better expression.

The full-length sequence of the polynucleotide of the present disclosure can usually be obtained through PCR amplification, recombination, or artificial synthesis. For the PCR amplification, primers can be designed according to the relevant nucleotide sequence disclosed in the present disclosure, especially the ORF sequence, and a commercially available cDNA library or a cDNA library prepared by a conventional method known to those skilled in the art can be used as a template to amplify the relevant sequence. When the sequence is long, it is often necessary to conduct two or more PCR amplifications, and then amplified fragments are spliced together in a correct order. The obtained nucleotide sequence can be cloned into a vector and then transformed into a cell, and then a large number of related sequences can be isolated from proliferated host cells by a conventional method. The mutation site of the present disclosure can also be introduced through artificial synthesis.

One or more copies of the polynucleotide of the present disclosure can be inserted into a host cell to increase a yield of a gene product. The copy number of the polynucleotide can be increased by integrating at least one additional copy of the sequence into a host cell genome or by integrating an amplifiable selectable marker gene with the polynucleotide, where in the latter case, a cell with the amplified copy of the selectable marker gene and the resulting additional copy of the polynucleotide can be selected by artificially cultivating the cell in the presence of a suitable selectable agent.

Methods well known to those skilled in the art can be used to construct a vector that includes a DNA sequence encoding the ACC mutant polypeptide and an appropriate transcription/translation control signal. The methods include in vitro recombinant DNA technology, DNA synthesis technology, and in vivo recombination technology. The DNA sequence

can be effectively linked to an appropriate promoter in a vector to guide mRNA synthesis. The vector may also include a ribosome binding site (RBS) for translation initiation and a transcription terminator.

The vectors applicable in the present disclosure may include commercially available plasmids, such as but not limited to: pBR322 (ATCC37017), pKK223-3 (Pharmacia Fine Chemicals, Uppsala, Sweden), GEM1 (Promega Biotech, Madison, Wis., USA), pQE70, pQE60, pQE-9 (Qiagen), pD10, psiX174pBluescript II KS, pNH8A, pNH16a, pNH18A, pNH46A (Stratagene), ptrc99a, pKK223-3, pKK233-3, pDR540, pRIT5 (Pharmacia), pKK232-8, pCM7, pSV2CAT, pOG44, pXT1, pSG (Stratagene), pSVK3, pBPV, pMSG, and pSVL (Pharmacia).

The present disclosure also provides a host cell carrying a nucleic acid sequence, a nucleic acid construct, or an expression vector encoding the ACC mutant polypeptide of the present disclosure. A vector carrying a nucleic acid encoding the protein of the present disclosure is introduced into a host cell, such that the vector exists as a part of a chromosomal integration or exists as a self-replicating extra-chromosomal vector described early, or the vector can achieve gene editing on the endogenous ACC gene of the host cell. The host cell may be any host cell familiar to those skilled in the art, including a prokaryotic cell and a eukaryotic cell.

The nucleic acid sequence, nucleic acid construct, or expression vector of the present disclosure can be introduced into a host cell through a variety of techniques, including transformation, transfection, transduction, viral infection, gene gun or Ti-plasmid-mediated gene delivery, calcium phosphate transfection, DEAE-dextran-mediated transfection, lipofection, electroporation, or the like.

In the production method of the present disclosure, the cells are cultivated on a nutrient medium suitable for the production of the polypeptide by a method well known in the art. If the polypeptide is secreted into the nutrient medium, the polypeptide can be directly recovered from the medium. If the polypeptide is not secreted into the medium, the polypeptide can be recovered from a cell lysate.

As used herein, the terms “gRNA”, “mature crRNA”, and “guide sequence” can be used interchangeably and have the meaning commonly understood by those skilled in the art. Generally, a gRNA can include direct repeats and guide sequences, or may be basically composed of or may be composed of direct repeats and guide sequences (also called spacers in the context of endogenous CRISPR systems).

In some cases, the guide sequence can be any polynucleotide sequence that shows sufficient complementarity with a target sequence to hybridize with the target sequence and guide the specific binding of the CRISPR/Cas complex to the target sequence. In an embodiment, under optimal alignment, a complementarity degree between the guide sequence and a corresponding target sequence may be at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, or at least 99%. Determining the optimal alignment is within the competence of those of ordinary skill in the art. For example, there are published and commercially available alignment algorithms and programs, including but not limited to Smith-Waterman, Bowtie, Geneious, Biopython, and SeqMan in ClustalW and matlab.

Sequence Listing

Sequence No.	Type	Content
5 SEQ ID NO: 1	Protein	Amino acid sequence of wild-type ACC
SEQ ID NO: 2	Protein	Amino acid sequence of I1879V
SEQ ID NO: 3	Protein	Amino acid sequence of C2186R
SEQ ID NO: 4	Protein	Amino acid sequence of I1879V and C2186R double-mutation
SEQ ID NO: 5	DNA	DNA sequence of I1879V
10 SEQ ID NO: 6	DNA	DNA sequence of C2186R
SEQ ID NO: 7	DNA	DNA sequence of I1879V and C2186R double-mutation

The present disclosure has the following advantages:

1. The present disclosure screens out a group of mutant ACC proteins.
2. A plant with the mutant ACC protein of the present disclosure shows obvious herbicide resistance compared with the wild-type plant.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic diagram of an ABE-nCas9 base editor, where OsU6 and ZmUbi are promoters; sgRNA is a gRNA; bp-NLS is a nuclear localization signal; and NOS is a terminator.

FIG. 2 shows the herbicide resistance of a plant with mutation of amino acid 1,879 of ACC relative to the wild-type.

FIG. 3 shows the herbicide resistance of a plant with mutation of amino acid 2,186 of ACC relative to the wild-type.

FIG. 4 shows the tolerance of an I1879V edited plant to sethoxydim.

FIG. 5 shows the tolerance of an I1879V edited plant to phentodim after phentodim is sprayed for 20 d.

FIG. 6 shows the tolerance of I1879V and C2186R edited plants to quizalofop-p and fenoxaprop-p-ethyl.

FIG. 7 shows the tolerance of an I1879V edited plant to different herbicides.

DETAILED DESCRIPTION OF THE EMBODIMENTS

The present disclosure will be further explained below in conjunction with examples. The following examples are only preferred examples of the present disclosure, and are not intended to limit the present disclosure in other forms. Any technical personnel familiar with the profession may use the technical content disclosed above to derive equivalent examples through equivalent changes. Any simple modification or equivalent change made to the following examples according to the technical essence of the present disclosure without departing from the content of the solutions of the present disclosure shall fall within the protection scope of the present disclosure.

Example 1 Construction of a Gene Editing Vector and Screening of a Mutation Site

1. An ABE-nCas9 base editor targeting the *Oryza sativa* endogenous ACC gene was constructed.

The ABE base editor could realize A/T->G/C base conversion within a specified sequence window. In the present disclosure, the ABE-nCas9 base editor was used as a vector, and a sgRNA (sgRNA shown in Table 1) was designed in the *Oryza sativa* endogenous ACC gene and cloned into the

ABE-nCas9 vector to form a base editor targeting the *Oryza sativa* endogenous ACC gene. An amino acid sequence encoded by the *Oryza sativa* endogenous ACC gene was shown in SEQ ID No: 1.

TABLE 1

sgRNA sequences (guide sequences) targeting <i>Oryza sativa</i> ACC	
sgRNA No.	guide-PAM sequence (5'-3')
A-ACC1879	GAGAATATACATGGAAGTGC
A-ACC2186	ATAGCACTCAATGCGGTCTG

2. *Oryza sativa* Genetic Transformation and Transgenic Plant Identification

Multiple *Oryza sativa* varieties such as Wanzhijing 006, Nanjing 9108, and Jiahe 218 were used as experimental materials. The base editor constructed above was transformed into the *Oryza sativa* plants by *A. tumefaciens* to obtain gene-edited plants. Emerging seedlings were screened with a sethoxydim-containing medium at a concentration of SET-2: 2 mg/L. Or, edited seedlings were planted in a cultivation room. The edited seedlings and the control plants were sprayed with sethoxydim at a concentration of 0.5 g/L (corresponding to a field application dosage of 10 g. a.i./mu), and a survival rate of the seedlings was counted 10 d later.

Through screening cultivation, herbicide-resistant plants were screened out on a sethoxydim-containing medium. As shown in FIG. 2, plants indicated by the arrow were able to tolerate the herbicide and showed resistance to the herbicide compared with other plants. The above herbicide-resistant plants were identified by PCR and sequencing, and it was found that the herbicide-resistant plants had an expected base substitution within a target range and a specific editing type was I1879V. In addition, herbicide-resistant plants were also screened out in the cultivation room. As shown in FIG. 3, edited plants were able to tolerate the herbicide and showed resistance to the herbicide compared with the control plants. The above herbicide-resistant plants were identified by PCR and sequencing, and it was found that the herbicide-resistant plants had an expected base substitution within a target range and a specific editing type was C2186R.

For I1879V edited *Oryza sativa* plants, the dosage of sethoxydim was further increased to a corresponding field application dosage of 15 g. a.i./mu. 14 d later, the control was obviously yellow and dead, and the edited material grew normally, as shown in FIG. 4. The dosage of sethoxydim was further increased to 30 g. a.i./mu. After the plants were further cultivated for 20 d, all wild-type plants died, but the edited plants still showed significant herbicide resistance, with phytotoxicity only of about 30%.

Example 2 Resistance to Other Types of Herbicides

The I1879V and C2186R edited *Oryza sativa* plants obtained in Example 1 were planted, and the resistance to

other types of herbicides was tested, such as phentodim, quizalofop-p, fenoxaprop-p-ethyl, clethodim, haloxyfop-p-methyl, fenoxaprop, and butyrclofen.

2.1 Phentodim:

Dibble *Oryza sativa* plants at the seedling stage in the field were selected and sprayed with phentodim at 20 g.a.i./mu (4 times a recommended dosage), and then results were observed. After the phentodim was sprayed for 7 d, the wild-type plants showed obvious chlorosis and spots, but the I1879V edited plants showed no visible phytotoxicity. After the phentodim was applied for 20 d, as shown in FIG. 5, the wild-type plants showed serious phytotoxicity (the arrow in FIG. 5 indicated obvious phytotoxicity), and its growth was also severely affected; and the I1879V edited plants showed little difference in biomass from the control group. That is, even 20 d later, the I1879V edited plants still showed excellent resistance to phentodim, had no obvious phytotoxicity, and were not affected in growth. However, compared with the wild-type control, the C2186R edited plant showed some resistance. Compared with the I1879V edited plant, the C2186R edited plant showed obvious phytotoxicity at 20 g.a.i./mu of phentodim, and there were chlorosis and spots at bases of leaves of the edited material, which was not safe enough for practical use.

2.2 Quizalofop-p and fenoxaprop-p-ethyl

Quizalofop-p and fenoxaprop-p-ethyl were each applied at 5 g.a.i./mu to I1879V edited plants and C2186R edited plants, and results were observed 14 d later. As shown in FIG. 6, 14 d later, there was no significant difference between the edited plants and the wild-type control, indicating that the above-mentioned edited plants failed to produce tolerance to quizalofop-p or fenoxaprop-p-ethyl.

2.3 Clethodim, haloxyfop-p-methyl, fenoxaprop, and butyrclofen:

The tolerance of the I1879V edited plants to clethodim, haloxyfop-p-methyl, fenoxaprop, and butyrclofen was tested, where the clethodim, haloxyfop-p-methyl, and fenoxaprop were each applied at concentrations of 5 g.a.i./mu (1×) and 10 g.a.i./mu (2×); and the butyrclofen was applied at concentrations of 10 g.a.i./mu (1×) and 20 g.a.i./mu (2×). After the herbicide was applied for 20 d, the growth of the edited plants was observed and the herbicide resistance of the edited plants was counted. As shown in FIG. 7, the I1879V edited plants showed some phytotoxicity at the above-mentioned 1× herbicide concentration, with a mortality rate of 30% to 50%; and at the above-mentioned 2× herbicide concentration, the edited plants were basically in a non-resistant state, with a mortality rate of 70% to 90%.

All documents mentioned in the present disclosure are cited as references in the present application, as if each document was individually cited as a reference. In addition, it should be understood that various changes or modifications may be made to the present disclosure by those skilled in the art after reading the above teaching content of the present disclosure, and these equivalent forms also fall within the scope defined by the appended claims of the present disclosure.

SEQUENCE LISTING

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-continued

<212> TYPE: PRT

<213> ORGANISM: *Oryza sativa*

<400> SEQUENCE: 1

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Ile Ser Pro Pro Pro Leu Gly Ile Ser Ala Phe Ser Arg Pro Leu Arg
          20           25           30

Leu Arg Leu Arg Leu Arg Leu Ala Ala Val Val Asp Ala Ala Arg Ile
          35           40           45

Gln Thr Pro Pro Arg Val Ala Phe Ser Ala Phe Phe Phe Phe Phe
50           55           60

Phe Phe Ser Glu Gly Leu Ser Arg Glu Gly Thr Ala Ala Arg Leu Pro
65           70           75           80

Ala Leu Ile Tyr Arg Val Pro Phe Ile Pro Cys Leu Leu Gln Leu Gln
          85           90           95

Ile Arg Ala Pro His Ala Asp Pro Arg Pro Ala Gly Phe Ser Cys Ala
          100          105          110

His His Phe Leu Ala Leu Ala Gln Gly Leu Ser Ser Ser Leu Leu Ser
          115          120          125

Gly Thr Ile Ser His Ile Trp Gly Phe Ile Phe Ser Leu Tyr Gly Thr
          130          135          140

Thr His Leu Arg Asn Arg Ala Ile Leu Leu Phe Gly Leu Ala Gly Ile
          145          150          155          160

Ile Asp Leu Pro Asn Asp Ala Ala Ser Glu Val Asp Ile Ser His Gly
          165          170          175

Ser Glu Asp Pro Arg Gly Pro Thr Val Pro Gly Ser Tyr Gln Met Asn
          180          185          190

Gly Ile Ile Asn Glu Thr His Asn Gly Arg His Ala Ser Val Ser Lys
          195          200          205

Val Val Glu Phe Cys Thr Ala Leu Gly Gly Lys Thr Pro Ile His Ser
          210          215          220

Val Leu Val Ala Asn Asn Gly Met Ala Ala Ala Lys Phe Met Arg Ser
          225          230          235          240

Val Arg Thr Trp Ala Asn Asp Thr Phe Gly Ser Glu Lys Ala Ile Gln
          245          250          255

Leu Ile Ala Met Ala Thr Pro Glu Asp Leu Arg Ile Asn Ala Glu His
          260          265          270

Ile Arg Ile Ala Asp Gln Phe Val Glu Val Pro Gly Gly Thr Asn Asn
          275          280          285

Asn Asn Tyr Ala Asn Val Gln Leu Ile Val Glu Ile Ala Glu Arg Thr
          290          295          300

Gly Val Ser Ala Val Trp Pro Gly Trp Gly His Ala Ser Glu Asn Pro
          305          310          315          320

Glu Leu Pro Asp Ala Leu Thr Ala Lys Gly Ile Val Phe Leu Gly Pro
          325          330          335

Pro Ala Ser Ser Met His Ala Leu Gly Asp Lys Val Gly Ser Ala Leu
          340          345          350

Ile Ala Gln Ala Ala Gly Val Pro Thr Leu Ala Trp Ser Gly Ser His
          355          360          365

Val Glu Val Pro Leu Glu Cys Cys Leu Asp Ser Ile Pro Asp Glu Met
          370          375          380

Tyr Arg Lys Ala Cys Val Thr Thr Thr Glu Glu Ala Val Ala Ser Cys
          385          390          395          400

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Gln Val Val Gly Tyr Pro Ala Met Ile Lys Ala Ser Trp Gly Gly Gly
 405 410 415
 Gly Lys Gly Ile Arg Lys Val His Asn Asp Asp Glu Val Arg Thr Leu
 420 425 430
 Phe Lys Gln Val Gln Gly Glu Val Pro Gly Ser Pro Ile Phe Ile Met
 435 440 445
 Arg Leu Ala Ala Gln Ser Arg His Leu Glu Val Gln Leu Leu Cys Asp
 450 455 460
 Gln Tyr Gly Asn Val Ala Ala Leu His Ser Arg Asp Cys Ser Val Gln
 465 470 475 480
 Arg Arg His Gln Lys Ile Ile Glu Glu Gly Pro Val Thr Val Ala Pro
 485 490 495
 Arg Glu Thr Val Lys Glu Leu Glu Gln Ala Ala Arg Arg Leu Ala Lys
 500 505 510
 Ala Val Gly Tyr Val Gly Ala Ala Thr Val Glu Tyr Leu Tyr Ser Met
 515 520 525
 Glu Thr Gly Glu Tyr Tyr Phe Leu Glu Leu Asn Pro Arg Leu Gln Val
 530 535 540
 Glu His Pro Val Thr Glu Trp Ile Ala Glu Val Asn Leu Pro Ala Ala
 545 550 555 560
 Gln Val Ala Val Gly Met Gly Ile Pro Leu Trp Gln Ile Pro Glu Ile
 565 570 575
 Arg Arg Phe Tyr Gly Met Asn His Gly Gly Gly Tyr Asp Leu Trp Arg
 580 585 590
 Lys Thr Ala Ala Leu Ala Thr Pro Phe Asn Phe Asp Glu Val Asp Ser
 595 600 605
 Lys Trp Pro Lys Gly His Cys Val Ala Val Arg Ile Thr Ser Glu Asp
 610 615 620
 Pro Asp Asp Gly Phe Lys Pro Thr Gly Gly Lys Val Lys Glu Ile Ser
 625 630 635 640
 Phe Lys Ser Lys Pro Asn Val Trp Ala Tyr Phe Ser Val Lys Ser Gly
 645 650 655
 Gly Gly Ile His Glu Phe Ala Asp Ser Gln Phe Gly His Val Phe Ala
 660 665 670
 Tyr Gly Thr Thr Arg Ser Ala Ala Ile Thr Thr Met Ala Leu Ala Leu
 675 680 685
 Lys Glu Val Gln Ile Arg Gly Glu Ile His Ser Asn Val Asp Tyr Thr
 690 695 700
 Val Asp Leu Leu Asn Ala Ser Asp Phe Arg Glu Asn Lys Ile His Thr
 705 710 715 720
 Gly Trp Leu Asp Thr Arg Ile Ala Met Arg Val Gln Ala Glu Arg Pro
 725 730 735
 Pro Trp Tyr Ile Ser Val Val Gly Gly Ala Leu Tyr Lys Thr Val Thr
 740 745 750
 Ala Asn Thr Ala Thr Val Ser Asp Tyr Val Gly Tyr Leu Thr Lys Gly
 755 760 765
 Gln Ile Pro Pro Lys His Ile Ser Leu Val Tyr Thr Thr Val Ala Leu
 770 775 780
 Asn Ile Asp Gly Lys Lys Tyr Thr Ile Asp Thr Val Arg Ser Gly His
 785 790 795 800
 Gly Ser Tyr Arg Leu Arg Met Asn Gly Ser Thr Val Asp Ala Asn Val
 805 810 815

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Gln Ile Leu Cys Asp Gly Gly Leu Leu Met Gln Leu Asp Gly Asn Ser	820	825	830
His Val Ile Tyr Ala Glu Glu Glu Ala Ser Gly Thr Arg Leu Leu Ile	835	840	845
Asp Gly Lys Thr Cys Met Leu Gln Asn Asp His Asp Pro Ser Lys Leu	850	855	860
Leu Ala Glu Thr Pro Cys Lys Leu Leu Arg Phe Leu Val Ala Asp Gly	865	870	875
Ala His Val Asp Ala Asp Val Pro Tyr Ala Glu Val Glu Val Met Lys	885	890	895
Met Cys Met Pro Leu Leu Ser Pro Ala Ser Gly Val Ile His Val Val	900	905	910
Met Ser Glu Gly Gln Ala Met Gln Ala Gly Asp Leu Ile Ala Arg Leu	915	920	925
Asp Leu Asp Asp Pro Ser Ala Val Lys Arg Ala Glu Pro Phe Glu Asp	930	935	940
Thr Phe Pro Gln Met Gly Leu Pro Ile Ala Ala Ser Gly Gln Val His	945	950	955
Lys Leu Cys Ala Ala Ser Leu Asn Ala Cys Arg Met Ile Leu Ala Gly	965	970	975
Tyr Glu His Asp Ile Asp Lys Val Val Pro Glu Leu Val Tyr Cys Leu	980	985	990
Asp Thr Pro Glu Leu Pro Phe Leu Gln Trp Glu Glu Leu Met Ser Val	995	1000	1005
Leu Ala Thr Arg Leu Pro Arg Asn Leu Lys Ser Glu Leu Glu Gly	1010	1015	1020
Lys Tyr Glu Glu Tyr Lys Val Lys Phe Asp Ser Gly Ile Ile Asn	1025	1030	1035
Asp Phe Pro Ala Asn Met Leu Arg Val Ile Ile Glu Glu Asn Leu	1040	1045	1050
Ala Cys Gly Ser Glu Lys Glu Lys Ala Thr Asn Glu Arg Leu Val	1055	1060	1065
Glu Pro Leu Met Ser Leu Leu Lys Ser Tyr Glu Gly Gly Arg Glu	1070	1075	1080
Ser His Ala His Phe Val Val Lys Ser Leu Phe Glu Glu Tyr Leu	1085	1090	1095
Tyr Val Glu Glu Leu Phe Ser Asp Gly Ile Gln Ser Asp Val Ile	1100	1105	1110
Glu Arg Leu Arg Leu Gln His Ser Lys Asp Leu Gln Lys Val Val	1115	1120	1125
Asp Ile Val Leu Ser His Gln Ser Val Arg Asn Lys Thr Lys Leu	1130	1135	1140
Ile Leu Lys Leu Met Glu Ser Leu Val Tyr Pro Asn Pro Ala Ala	1145	1150	1155
Tyr Arg Asp Gln Leu Ile Arg Phe Ser Ser Leu Asn His Lys Ala	1160	1165	1170
Tyr Tyr Lys Leu Ala Leu Lys Ala Ser Glu Leu Leu Glu Gln Thr	1175	1180	1185
Lys Leu Ser Glu Leu Arg Ala Arg Ile Ala Arg Ser Leu Ser Glu	1190	1195	1200
Leu Glu Met Phe Thr Glu Glu Ser Lys Gly Leu Ser Met His Lys	1205	1210	1215
Arg Glu Ile Ala Ile Lys Glu Ser Met Glu Asp Leu Val Thr Ala			

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1220	1225	1230
Pro Leu Pro Val Glu Asp Ala Leu Ile Ser Leu Phe Asp Cys Ser		
1235	1240	1245
Asp Thr Thr Val Gln Gln Arg Val Ile Glu Thr Tyr Ile Ala Arg		
1250	1255	1260
Leu Tyr Gln Pro His Leu Val Lys Asp Ser Ile Lys Met Lys Trp		
1265	1270	1275
Ile Glu Ser Gly Val Ile Ala Leu Trp Glu Phe Pro Glu Gly His		
1280	1285	1290
Phe Asp Ala Arg Asn Gly Gly Ala Val Leu Gly Asp Lys Arg Trp		
1295	1300	1305
Gly Ala Met Val Ile Val Lys Ser Leu Glu Ser Leu Ser Met Ala		
1310	1315	1320
Ile Arg Phe Ala Leu Lys Glu Thr Ser His Tyr Thr Ser Ser Glu		
1325	1330	1335
Gly Asn Met Met His Ile Ala Leu Leu Gly Ala Asp Asn Lys Met		
1340	1345	1350
His Ile Ile Gln Glu Ser Gly Asp Asp Ala Asp Arg Ile Ala Lys		
1355	1360	1365
Leu Pro Leu Ile Leu Lys Asp Asn Val Thr Asp Leu His Ala Ser		
1370	1375	1380
Gly Val Lys Thr Ile Ser Phe Ile Val Gln Arg Asp Glu Ala Arg		
1385	1390	1395
Met Thr Met Arg Arg Thr Phe Leu Trp Ser Asp Glu Lys Leu Ser		
1400	1405	1410
Tyr Glu Glu Glu Pro Ile Leu Arg His Val Glu Pro Pro Leu Ser		
1415	1420	1425
Ala Leu Leu Glu Leu Asp Lys Leu Lys Val Lys Gly Tyr Asn Glu		
1430	1435	1440
Met Lys Tyr Thr Pro Ser Arg Asp Arg Gln Trp His Ile Tyr Thr		
1445	1450	1455
Leu Arg Asn Thr Glu Asn Pro Lys Met Leu His Arg Val Phe Phe		
1460	1465	1470
Arg Thr Leu Val Arg Gln Pro Ser Val Ser Asn Lys Phe Ser Ser		
1475	1480	1485
Gly Gln Ile Gly Asp Met Glu Val Gly Ser Ala Glu Glu Pro Leu		
1490	1495	1500
Ser Phe Thr Ser Thr Ser Ile Leu Arg Ser Leu Met Thr Ala Ile		
1505	1510	1515
Glu Glu Leu Glu Leu His Ala Ile Arg Thr Gly His Ser His Met		
1520	1525	1530
Tyr Leu His Val Leu Lys Glu Gln Lys Leu Leu Asp Leu Val Pro		
1535	1540	1545
Val Ser Gly Asn Thr Val Leu Asp Val Gly Gln Asp Glu Ala Thr		
1550	1555	1560
Ala Tyr Ser Leu Leu Lys Glu Met Ala Met Lys Ile His Glu Leu		
1565	1570	1575
Val Gly Ala Arg Met His His Leu Ser Val Cys Gln Trp Glu Val		
1580	1585	1590
Lys Leu Lys Leu Asp Cys Asp Gly Pro Ala Ser Gly Thr Trp Arg		
1595	1600	1605
Ile Val Thr Thr Asn Val Thr Ser His Thr Cys Thr Val Asp Ile		
1610	1615	1620

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Tyr Arg	Glu Met	Glu Asp	Lys	Glu Ser	Arg Lys	Leu	Val Tyr	His	
1625			1630			1635			
Pro Ala	Thr Pro	Ala Ala	Gly	Pro Leu	His Gly	Val	Ala Leu	Asn	
1640			1645			1650			
Asn Pro	Tyr Gln	Pro Leu	Ser	Val Ile	Asp Leu	Lys	Arg Cys	Ser	
1655			1660			1665			
Ala Arg	Asn Asn	Arg Thr	Thr	Tyr Cys	Tyr Asp	Phe	Pro Leu	Ala	
1670			1675			1680			
Phe Glu	Thr Ala	Val Arg	Lys	Ser Trp	Ser Ser	Ser	Thr Ser	Gly	
1685			1690			1695			
Ala Ser	Lys Gly	Val Glu	Asn	Ala Gln	Cys Tyr	Val	Lys Ala	Thr	
1700			1705			1710			
Glu Leu	Val Phe	Ala Asp	Lys	His Gly	Ser Trp	Gly	Thr Pro	Leu	
1715			1720			1725			
Val Gln	Met Asp	Arg Pro	Ala	Gly Leu	Asn Asp	Ile	Gly Met	Val	
1730			1735			1740			
Ala Trp	Thr Leu	Lys Met	Ser	Thr Pro	Glu Phe	Pro	Ser Gly	Arg	
1745			1750			1755			
Glu Ile	Ile Val	Val Ala	Asn	Asp Ile	Thr Phe	Arg	Ala Gly	Ser	
1760			1765			1770			
Phe Gly	Pro Arg	Glu Asp	Ala	Phe Phe	Glu Ala	Val	Thr Asn	Leu	
1775			1780			1785			
Ala Cys	Glu Lys	Lys Leu	Pro	Leu Ile	Tyr Leu	Ala	Ala Asn	Ser	
1790			1795			1800			
Gly Ala	Arg Ile	Gly Ile	Ala	Asp Glu	Val Lys	Ser	Cys Phe	Arg	
1805			1810			1815			
Val Gly	Trp Ser	Asp Asp	Gly	Ser Pro	Glu Arg	Gly	Phe Gln	Tyr	
1820			1825			1830			
Ile Tyr	Leu Ser	Glu Glu	Asp	Tyr Ala	Arg Ile	Gly	Thr Ser	Val	
1835			1840			1845			
Ile Ala	His Lys	Met Gln	Leu	Asp Ser	Gly Glu	Ile	Arg Trp	Val	
1850			1855			1860			
Ile Asp	Ser Val	Val Gly	Lys	Glu Asp	Gly Leu	Gly	Val Glu	Asn	
1865			1870			1875			
Ile His	Gly Ser	Ala Ala	Ile	Ala Ser	Ala Tyr	Ser	Arg Ala	Tyr	
1880			1885			1890			
Lys Glu	Thr Phe	Thr Leu	Thr	Phe Val	Thr Gly	Arg	Thr Val	Gly	
1895			1900			1905			
Ile Gly	Ala Tyr	Leu Ala	Arg	Leu Gly	Ile Arg	Cys	Ile Gln	Arg	
1910			1915			1920			
Leu Asp	Gln Pro	Ile Ile	Leu	Thr Gly	Tyr Ser	Ala	Leu Asn	Lys	
1925			1930			1935			
Leu Leu	Gly Arg	Glu Val	Tyr	Ser Ser	His Met	Gln	Leu Gly	Gly	
1940			1945			1950			
Pro Lys	Ile Met	Ala Thr	Asn	Gly Val	Val His	Leu	Thr Val	Ser	
1955			1960			1965			
Asp Asp	Leu Glu	Gly Val	Ser	Asn Ile	Leu Arg	Trp	Leu Ser	Tyr	
1970			1975			1980			
Val Pro	Ala Tyr	Ile Gly	Gly	Pro Leu	Pro Val	Thr	Thr Pro	Leu	
1985			1990			1995			
Asp Pro	Pro Asp	Arg Pro	Val	Ala Tyr	Ile Pro	Glu	Asn Ser	Cys	
2000			2005			2010			

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Asp 2015	Pro	Arg	Ala	Ala	Ile	Arg 2020	Gly	Val	Asp	Asp	Ser 2025	Gln	Gly	Lys
Trp 2030	Leu	Gly	Gly	Met	Phe	Asp 2035	Lys	Asp	Ser	Phe	Val 2040	Glu	Thr	Phe
Glu 2045	Gly	Trp	Ala	Lys	Thr	Val 2050	Val	Thr	Gly	Arg	Ala 2055	Lys	Leu	Gly
Gly 2060	Ile	Pro	Val	Gly	Val	Ile 2065	Ala	Val	Glu	Thr	Gln 2070	Thr	Met	Met
Gln 2075	Thr	Ile	Pro	Ala	Asp	Pro 2080	Gly	Gln	Leu	Asp	Ser 2085	Arg	Glu	Gln
Ser 2090	Val	Pro	Arg	Ala	Gly	Gln 2095	Val	Trp	Phe	Pro	Asp 2100	Ser	Ala	Thr
Lys 2105	Thr	Ala	Gln	Ala	Leu	Leu 2110	Asp	Phe	Asn	Arg	Glu 2115	Gly	Leu	Pro
Leu 2120	Phe	Ile	Leu	Ala	Asn	Trp 2125	Arg	Gly	Phe	Ser	Gly 2130	Gly	Gln	Arg
Asp 2135	Leu	Phe	Glu	Gly	Ile	Leu 2140	Gln	Ala	Gly	Ser	Thr 2145	Ile	Val	Glu
Asn 2150	Leu	Arg	Thr	Tyr	Asn	Gln 2155	Pro	Ala	Phe	Val	Tyr 2160	Ile	Pro	Met
Ala 2165	Ala	Glu	Leu	Arg	Gly	Gly 2170	Ala	Trp	Val	Val	Val 2175	Asp	Ser	Lys
Ile 2180	Asn	Pro	Asp	Arg	Ile	Glu 2185	Cys	Tyr	Ala	Glu	Arg 2190	Thr	Ala	Lys
Gly 2195	Asn	Val	Leu	Glu	Pro	Gln 2200	Gly	Leu	Ile	Glu	Ile 2205	Lys	Phe	Arg
Ser 2210	Glu	Glu	Leu	Gln	Asp	Cys 2215	Met	Ser	Arg	Leu	Asp 2220	Pro	Thr	Leu
Ile 2225	Asp	Leu	Lys	Ala	Lys	Leu 2230	Glu	Val	Ala	Asn	Lys 2235	Asn	Gly	Ser
Ala 2240	Asp	Thr	Lys	Ser	Leu	Gln 2245	Glu	Asn	Ile	Glu	Ala 2250	Arg	Thr	Lys
Gln 2255	Leu	Met	Pro	Leu	Tyr	Thr 2260	Gln	Ile	Ala	Ile	Arg 2265	Phe	Ala	Glu
Leu 2270	His	Asp	Thr	Ser	Leu	Arg 2275	Met	Ala	Ala	Lys	Gly 2280	Val	Ile	Lys
Lys 2285	Val	Val	Asp	Trp	Glu	Glu 2290	Ser	Arg	Ser	Phe	Phe 2295	Tyr	Lys	Arg
Leu 2300	Arg	Arg	Arg	Ile	Ser	Glu 2305	Asp	Val	Leu	Ala	Lys 2310	Glu	Ile	Arg
Ala 2315	Val	Ala	Gly	Glu	Gln	Phe 2320	Ser	His	Gln	Pro	Ala 2325	Ile	Glu	Leu
Ile 2330	Lys	Lys	Trp	Tyr	Ser	Ala 2335	Ser	His	Ala	Ala	Glu 2340	Trp	Asp	Asp
Asp 2345	Asp	Ala	Phe	Val	Ala	Trp 2350	Met	Asp	Asn	Pro	Glu 2355	Asn	Tyr	Lys
Asp 2360	Tyr	Ile	Gln	Tyr	Leu	Lys 2365	Ala	Gln	Arg	Val	Ser 2370	Gln	Ser	Leu
Ser 2375	Ser	Leu	Ser	Asp	Ser	Ser 2380	Ser	Asp	Leu	Gln	Ala 2385	Leu	Pro	Gln
Gly 2390	Leu	Ser	Met	Leu	Leu	Asp 2395	Lys	Met	Asp	Pro	Ser 2400	Arg	Arg	Ala
Gln 2405	Leu	Val	Glu	Glu	Ile	Arg	Lys	Val	Leu	Gly				

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2405 2410

<210> SEQ ID NO 2
 <211> LENGTH: 2414
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: mutant ACC

<400> SEQUENCE: 2

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Ile Ser Pro Pro Pro Leu Gly Ile Ser Ala Phe Ser Arg Pro Leu Arg
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Leu Arg Leu Arg Leu Arg Leu Ala Ala Val Val Asp Ala Ala Arg Ile
 35 40 45

Gln Thr Pro Pro Arg Val Ala Phe Ser Ala Phe Phe Phe Phe Phe
 50 55 60

Phe Phe Ser Glu Gly Leu Ser Arg Glu Gly Thr Ala Ala Arg Leu Pro
 65 70 75 80

Ala Leu Ile Tyr Arg Val Pro Phe Ile Pro Cys Leu Leu Gln Leu Gln
 85 90 95

Ile Arg Ala Pro His Ala Asp Pro Arg Pro Ala Gly Phe Ser Cys Ala
 100 105 110

His His Phe Leu Ala Leu Ala Gln Gly Leu Ser Ser Ser Leu Leu Ser
 115 120 125

Gly Thr Ile Ser His Ile Trp Gly Phe Ile Phe Ser Leu Tyr Gly Thr
 130 135 140

Thr His Leu Arg Asn Arg Ala Ile Leu Leu Phe Gly Leu Ala Gly Ile
 145 150 155 160

Ile Asp Leu Pro Asn Asp Ala Ala Ser Glu Val Asp Ile Ser His Gly
 165 170 175

Ser Glu Asp Pro Arg Gly Pro Thr Val Pro Gly Ser Tyr Gln Met Asn
 180 185 190

Gly Ile Ile Asn Glu Thr His Asn Gly Arg His Ala Ser Val Ser Lys
 195 200 205

Val Val Glu Phe Cys Thr Ala Leu Gly Gly Lys Thr Pro Ile His Ser
 210 215 220

Val Leu Val Ala Asn Asn Gly Met Ala Ala Ala Lys Phe Met Arg Ser
 225 230 235 240

Val Arg Thr Trp Ala Asn Asp Thr Phe Gly Ser Glu Lys Ala Ile Gln
 245 250 255

Leu Ile Ala Met Ala Thr Pro Glu Asp Leu Arg Ile Asn Ala Glu His
 260 265 270

Ile Arg Ile Ala Asp Gln Phe Val Glu Val Pro Gly Gly Thr Asn Asn
 275 280 285

Asn Asn Tyr Ala Asn Val Gln Leu Ile Val Glu Ile Ala Glu Arg Thr
 290 295 300

Gly Val Ser Ala Val Trp Pro Gly Trp Gly His Ala Ser Glu Asn Pro
 305 310 315 320

Glu Leu Pro Asp Ala Leu Thr Ala Lys Gly Ile Val Phe Leu Gly Pro
 325 330 335

Pro Ala Ser Ser Met His Ala Leu Gly Asp Lys Val Gly Ser Ala Leu
 340 345 350

Ile Ala Gln Ala Ala Gly Val Pro Thr Leu Ala Trp Ser Gly Ser His

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355					360					365					
Val	Glu	Val	Pro	Leu	Glu	Cys	Cys	Leu	Asp	Ser	Ile	Pro	Asp	Glu	Met
370						375					380				
Tyr	Arg	Lys	Ala	Cys	Val	Thr	Thr	Thr	Glu	Glu	Ala	Val	Ala	Ser	Cys
385					390					395					400
Gln	Val	Val	Gly	Tyr	Pro	Ala	Met	Ile	Lys	Ala	Ser	Trp	Gly	Gly	Gly
				405					410					415	
Gly	Lys	Gly	Ile	Arg	Lys	Val	His	Asn	Asp	Asp	Glu	Val	Arg	Thr	Leu
			420					425					430		
Phe	Lys	Gln	Val	Gln	Gly	Glu	Val	Pro	Gly	Ser	Pro	Ile	Phe	Ile	Met
		435					440					445			
Arg	Leu	Ala	Ala	Gln	Ser	Arg	His	Leu	Glu	Val	Gln	Leu	Leu	Cys	Asp
	450					455					460				
Gln	Tyr	Gly	Asn	Val	Ala	Ala	Leu	His	Ser	Arg	Asp	Cys	Ser	Val	Gln
465				470						475					480
Arg	Arg	His	Gln	Lys	Ile	Ile	Glu	Glu	Gly	Pro	Val	Thr	Val	Ala	Pro
				485					490					495	
Arg	Glu	Thr	Val	Lys	Glu	Leu	Glu	Gln	Ala	Ala	Arg	Arg	Leu	Ala	Lys
			500					505					510		
Ala	Val	Gly	Tyr	Val	Gly	Ala	Ala	Thr	Val	Glu	Tyr	Leu	Tyr	Ser	Met
		515					520					525			
Glu	Thr	Gly	Glu	Tyr	Tyr	Phe	Leu	Glu	Leu	Asn	Pro	Arg	Leu	Gln	Val
	530					535					540				
Glu	His	Pro	Val	Thr	Glu	Trp	Ile	Ala	Glu	Val	Asn	Leu	Pro	Ala	Ala
545					550					555					560
Gln	Val	Ala	Val	Gly	Met	Gly	Ile	Pro	Leu	Trp	Gln	Ile	Pro	Glu	Ile
				565					570					575	
Arg	Arg	Phe	Tyr	Gly	Met	Asn	His	Gly	Gly	Gly	Tyr	Asp	Leu	Trp	Arg
			580					585					590		
Lys	Thr	Ala	Ala	Leu	Ala	Thr	Pro	Phe	Asn	Phe	Asp	Glu	Val	Asp	Ser
		595					600					605			
Lys	Trp	Pro	Lys	Gly	His	Cys	Val	Ala	Val	Arg	Ile	Thr	Ser	Glu	Asp
	610					615					620				
Pro	Asp	Asp	Gly	Phe	Lys	Pro	Thr	Gly	Gly	Lys	Val	Lys	Glu	Ile	Ser
625					630					635					640
Phe	Lys	Ser	Lys	Pro	Asn	Val	Trp	Ala	Tyr	Phe	Ser	Val	Lys	Ser	Gly
				645					650					655	
Gly	Gly	Ile	His	Glu	Phe	Ala	Asp	Ser	Gln	Phe	Gly	His	Val	Phe	Ala
			660					665					670		
Tyr	Gly	Thr	Thr	Arg	Ser	Ala	Ala	Ile	Thr	Thr	Met	Ala	Leu	Ala	Leu
		675					680					685			
Lys	Glu	Val	Gln	Ile	Arg	Gly	Glu	Ile	His	Ser	Asn	Val	Asp	Tyr	Thr
	690					695					700				
Val	Asp	Leu	Leu	Asn	Ala	Ser	Asp	Phe	Arg	Glu	Asn	Lys	Ile	His	Thr
	705				710					715					720
Gly	Trp	Leu	Asp	Thr	Arg	Ile	Ala	Met	Arg	Val	Gln	Ala	Glu	Arg	Pro
				725					730					735	
Pro	Trp	Tyr	Ile	Ser	Val	Val	Gly	Gly	Ala	Leu	Tyr	Lys	Thr	Val	Thr
			740					745					750		
Ala	Asn	Thr	Ala	Thr	Val	Ser	Asp	Tyr	Val	Gly	Tyr	Leu	Thr	Lys	Gly
		755					760					765			
Gln	Ile	Pro	Pro	Lys	His	Ile	Ser	Leu	Val	Tyr	Thr	Thr	Val	Ala	Leu
	770					775						780			

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Asn Ile Asp Gly Lys Lys Tyr Thr Ile Asp Thr Val Arg Ser Gly His
 785 790 795 800
 Gly Ser Tyr Arg Leu Arg Met Asn Gly Ser Thr Val Asp Ala Asn Val
 805 810 815
 Gln Ile Leu Cys Asp Gly Gly Leu Leu Met Gln Leu Asp Gly Asn Ser
 820 825 830
 His Val Ile Tyr Ala Glu Glu Glu Ala Ser Gly Thr Arg Leu Leu Ile
 835 840 845
 Asp Gly Lys Thr Cys Met Leu Gln Asn Asp His Asp Pro Ser Lys Leu
 850 855 860
 Leu Ala Glu Thr Pro Cys Lys Leu Leu Arg Phe Leu Val Ala Asp Gly
 865 870 875 880
 Ala His Val Asp Ala Asp Val Pro Tyr Ala Glu Val Glu Val Met Lys
 885 890 895
 Met Cys Met Pro Leu Leu Ser Pro Ala Ser Gly Val Ile His Val Val
 900 905 910
 Met Ser Glu Gly Gln Ala Met Gln Ala Gly Asp Leu Ile Ala Arg Leu
 915 920 925
 Asp Leu Asp Asp Pro Ser Ala Val Lys Arg Ala Glu Pro Phe Glu Asp
 930 935 940
 Thr Phe Pro Gln Met Gly Leu Pro Ile Ala Ala Ser Gly Gln Val His
 945 950 955 960
 Lys Leu Cys Ala Ala Ser Leu Asn Ala Cys Arg Met Ile Leu Ala Gly
 965 970 975
 Tyr Glu His Asp Ile Asp Lys Val Val Pro Glu Leu Val Tyr Cys Leu
 980 985 990
 Asp Thr Pro Glu Leu Pro Phe Leu Gln Trp Glu Glu Leu Met Ser Val
 995 1000 1005
 Leu Ala Thr Arg Leu Pro Arg Asn Leu Lys Ser Glu Leu Glu Gly
 1010 1015 1020
 Lys Tyr Glu Glu Tyr Lys Val Lys Phe Asp Ser Gly Ile Ile Asn
 1025 1030 1035
 Asp Phe Pro Ala Asn Met Leu Arg Val Ile Ile Glu Glu Asn Leu
 1040 1045 1050
 Ala Cys Gly Ser Glu Lys Glu Lys Ala Thr Asn Glu Arg Leu Val
 1055 1060 1065
 Glu Pro Leu Met Ser Leu Leu Lys Ser Tyr Glu Gly Gly Arg Glu
 1070 1075 1080
 Ser His Ala His Phe Val Val Lys Ser Leu Phe Glu Glu Tyr Leu
 1085 1090 1095
 Tyr Val Glu Glu Leu Phe Ser Asp Gly Ile Gln Ser Asp Val Ile
 1100 1105 1110
 Glu Arg Leu Arg Leu Gln His Ser Lys Asp Leu Gln Lys Val Val
 1115 1120 1125
 Asp Ile Val Leu Ser His Gln Ser Val Arg Asn Lys Thr Lys Leu
 1130 1135 1140
 Ile Leu Lys Leu Met Glu Ser Leu Val Tyr Pro Asn Pro Ala Ala
 1145 1150 1155
 Tyr Arg Asp Gln Leu Ile Arg Phe Ser Ser Leu Asn His Lys Ala
 1160 1165 1170
 Tyr Tyr Lys Leu Ala Leu Lys Ala Ser Glu Leu Leu Glu Gln Thr
 1175 1180 1185

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Lys	Leu	Ser	Glu	Leu	Arg	Ala	Arg	Ile	Ala	Arg	Ser	Leu	Ser	Glu
1190						1195					1200			
Leu	Glu	Met	Phe	Thr	Glu	Glu	Ser	Lys	Gly	Leu	Ser	Met	His	Lys
1205						1210					1215			
Arg	Glu	Ile	Ala	Ile	Lys	Glu	Ser	Met	Glu	Asp	Leu	Val	Thr	Ala
1220						1225					1230			
Pro	Leu	Pro	Val	Glu	Asp	Ala	Leu	Ile	Ser	Leu	Phe	Asp	Cys	Ser
1235						1240					1245			
Asp	Thr	Thr	Val	Gln	Gln	Arg	Val	Ile	Glu	Thr	Tyr	Ile	Ala	Arg
1250						1255					1260			
Leu	Tyr	Gln	Pro	His	Leu	Val	Lys	Asp	Ser	Ile	Lys	Met	Lys	Trp
1265						1270					1275			
Ile	Glu	Ser	Gly	Val	Ile	Ala	Leu	Trp	Glu	Phe	Pro	Glu	Gly	His
1280						1285					1290			
Phe	Asp	Ala	Arg	Asn	Gly	Gly	Ala	Val	Leu	Gly	Asp	Lys	Arg	Trp
1295						1300					1305			
Gly	Ala	Met	Val	Ile	Val	Lys	Ser	Leu	Glu	Ser	Leu	Ser	Met	Ala
1310						1315					1320			
Ile	Arg	Phe	Ala	Leu	Lys	Glu	Thr	Ser	His	Tyr	Thr	Ser	Ser	Glu
1325						1330					1335			
Gly	Asn	Met	Met	His	Ile	Ala	Leu	Leu	Gly	Ala	Asp	Asn	Lys	Met
1340						1345					1350			
His	Ile	Ile	Gln	Glu	Ser	Gly	Asp	Asp	Ala	Asp	Arg	Ile	Ala	Lys
1355						1360					1365			
Leu	Pro	Leu	Ile	Leu	Lys	Asp	Asn	Val	Thr	Asp	Leu	His	Ala	Ser
1370						1375					1380			
Gly	Val	Lys	Thr	Ile	Ser	Phe	Ile	Val	Gln	Arg	Asp	Glu	Ala	Arg
1385						1390					1395			
Met	Thr	Met	Arg	Arg	Thr	Phe	Leu	Trp	Ser	Asp	Glu	Lys	Leu	Ser
1400						1405					1410			
Tyr	Glu	Glu	Glu	Pro	Ile	Leu	Arg	His	Val	Glu	Pro	Pro	Leu	Ser
1415						1420					1425			
Ala	Leu	Leu	Glu	Leu	Asp	Lys	Leu	Lys	Val	Lys	Gly	Tyr	Asn	Glu
1430						1435					1440			
Met	Lys	Tyr	Thr	Pro	Ser	Arg	Asp	Arg	Gln	Trp	His	Ile	Tyr	Thr
1445						1450					1455			
Leu	Arg	Asn	Thr	Glu	Asn	Pro	Lys	Met	Leu	His	Arg	Val	Phe	Phe
1460						1465					1470			
Arg	Thr	Leu	Val	Arg	Gln	Pro	Ser	Val	Ser	Asn	Lys	Phe	Ser	Ser
1475						1480					1485			
Gly	Gln	Ile	Gly	Asp	Met	Glu	Val	Gly	Ser	Ala	Glu	Glu	Pro	Leu
1490						1495					1500			
Ser	Phe	Thr	Ser	Thr	Ser	Ile	Leu	Arg	Ser	Leu	Met	Thr	Ala	Ile
1505						1510					1515			
Glu	Glu	Leu	Glu	Leu	His	Ala	Ile	Arg	Thr	Gly	His	Ser	His	Met
1520						1525					1530			
Tyr	Leu	His	Val	Leu	Lys	Glu	Gln	Lys	Leu	Leu	Asp	Leu	Val	Pro
1535						1540					1545			
Val	Ser	Gly	Asn	Thr	Val	Leu	Asp	Val	Gly	Gln	Asp	Glu	Ala	Thr
1550						1555					1560			
Ala	Tyr	Ser	Leu	Leu	Lys	Glu	Met	Ala	Met	Lys	Ile	His	Glu	Leu
1565						1570					1575			
Val	Gly	Ala	Arg	Met	His	His	Leu	Ser	Val	Cys	Gln	Trp	Glu	Val

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1580	1585	1590
Lys Leu Lys Leu Asp Cys Asp Gly Pro Ala Ser Gly Thr Trp Arg		
1595	1600	1605
Ile Val Thr Thr Asn Val Thr Ser His Thr Cys Thr Val Asp Ile		
1610	1615	1620
Tyr Arg Glu Met Glu Asp Lys Glu Ser Arg Lys Leu Val Tyr His		
1625	1630	1635
Pro Ala Thr Pro Ala Ala Gly Pro Leu His Gly Val Ala Leu Asn		
1640	1645	1650
Asn Pro Tyr Gln Pro Leu Ser Val Ile Asp Leu Lys Arg Cys Ser		
1655	1660	1665
Ala Arg Asn Asn Arg Thr Thr Tyr Cys Tyr Asp Phe Pro Leu Ala		
1670	1675	1680
Phe Glu Thr Ala Val Arg Lys Ser Trp Ser Ser Ser Thr Ser Gly		
1685	1690	1695
Ala Ser Lys Gly Val Glu Asn Ala Gln Cys Tyr Val Lys Ala Thr		
1700	1705	1710
Glu Leu Val Phe Ala Asp Lys His Gly Ser Trp Gly Thr Pro Leu		
1715	1720	1725
Val Gln Met Asp Arg Pro Ala Gly Leu Asn Asp Ile Gly Met Val		
1730	1735	1740
Ala Trp Thr Leu Lys Met Ser Thr Pro Glu Phe Pro Ser Gly Arg		
1745	1750	1755
Glu Ile Ile Val Val Ala Asn Asp Ile Thr Phe Arg Ala Gly Ser		
1760	1765	1770
Phe Gly Pro Arg Glu Asp Ala Phe Phe Glu Ala Val Thr Asn Leu		
1775	1780	1785
Ala Cys Glu Lys Lys Leu Pro Leu Ile Tyr Leu Ala Ala Asn Ser		
1790	1795	1800
Gly Ala Arg Ile Gly Ile Ala Asp Glu Val Lys Ser Cys Phe Arg		
1805	1810	1815
Val Gly Trp Ser Asp Asp Gly Ser Pro Glu Arg Gly Phe Gln Tyr		
1820	1825	1830
Ile Tyr Leu Ser Glu Glu Asp Tyr Ala Arg Ile Gly Thr Ser Val		
1835	1840	1845
Ile Ala His Lys Met Gln Leu Asp Ser Gly Glu Ile Arg Trp Val		
1850	1855	1860
Ile Asp Ser Val Val Gly Lys Glu Asp Gly Leu Gly Val Glu Asn		
1865	1870	1875
Val His Gly Ser Ala Ala Ile Ala Ser Ala Tyr Ser Arg Ala Tyr		
1880	1885	1890
Lys Glu Thr Phe Thr Leu Thr Phe Val Thr Gly Arg Thr Val Gly		
1895	1900	1905
Ile Gly Ala Tyr Leu Ala Arg Leu Gly Ile Arg Cys Ile Gln Arg		
1910	1915	1920
Leu Asp Gln Pro Ile Ile Leu Thr Gly Tyr Ser Ala Leu Asn Lys		
1925	1930	1935
Leu Leu Gly Arg Glu Val Tyr Ser Ser His Met Gln Leu Gly Gly		
1940	1945	1950
Pro Lys Ile Met Ala Thr Asn Gly Val Val His Leu Thr Val Ser		
1955	1960	1965
Asp Asp Leu Glu Gly Val Ser Asn Ile Leu Arg Trp Leu Ser Tyr		
1970	1975	1980

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Val	Pro	Ala	Tyr	Ile	Gly	Gly	Pro	Leu	Pro	Val	Thr	Thr	Pro	Leu
1985						1990					1995			
Asp	Pro	Pro	Asp	Arg	Pro	Val	Ala	Tyr	Ile	Pro	Glu	Asn	Ser	Cys
2000						2005					2010			
Asp	Pro	Arg	Ala	Ala	Ile	Arg	Gly	Val	Asp	Asp	Ser	Gln	Gly	Lys
2015						2020					2025			
Trp	Leu	Gly	Gly	Met	Phe	Asp	Lys	Asp	Ser	Phe	Val	Glu	Thr	Phe
2030						2035					2040			
Glu	Gly	Trp	Ala	Lys	Thr	Val	Val	Thr	Gly	Arg	Ala	Lys	Leu	Gly
2045						2050					2055			
Gly	Ile	Pro	Val	Gly	Val	Ile	Ala	Val	Glu	Thr	Gln	Thr	Met	Met
2060						2065					2070			
Gln	Thr	Ile	Pro	Ala	Asp	Pro	Gly	Gln	Leu	Asp	Ser	Arg	Glu	Gln
2075						2080					2085			
Ser	Val	Pro	Arg	Ala	Gly	Gln	Val	Trp	Phe	Pro	Asp	Ser	Ala	Thr
2090						2095					2100			
Lys	Thr	Ala	Gln	Ala	Leu	Leu	Asp	Phe	Asn	Arg	Glu	Gly	Leu	Pro
2105						2110					2115			
Leu	Phe	Ile	Leu	Ala	Asn	Trp	Arg	Gly	Phe	Ser	Gly	Gly	Gln	Arg
2120						2125					2130			
Asp	Leu	Phe	Glu	Gly	Ile	Leu	Gln	Ala	Gly	Ser	Thr	Ile	Val	Glu
2135						2140					2145			
Asn	Leu	Arg	Thr	Tyr	Asn	Gln	Pro	Ala	Phe	Val	Tyr	Ile	Pro	Met
2150						2155					2160			
Ala	Ala	Glu	Leu	Arg	Gly	Gly	Ala	Trp	Val	Val	Val	Asp	Ser	Lys
2165						2170					2175			
Ile	Asn	Pro	Asp	Arg	Ile	Glu	Cys	Tyr	Ala	Glu	Arg	Thr	Ala	Lys
2180						2185					2190			
Gly	Asn	Val	Leu	Glu	Pro	Gln	Gly	Leu	Ile	Glu	Ile	Lys	Phe	Arg
2195						2200					2205			
Ser	Glu	Glu	Leu	Gln	Asp	Cys	Met	Ser	Arg	Leu	Asp	Pro	Thr	Leu
2210						2215					2220			
Ile	Asp	Leu	Lys	Ala	Lys	Leu	Glu	Val	Ala	Asn	Lys	Asn	Gly	Ser
2225						2230					2235			
Ala	Asp	Thr	Lys	Ser	Leu	Gln	Glu	Asn	Ile	Glu	Ala	Arg	Thr	Lys
2240						2245					2250			
Gln	Leu	Met	Pro	Leu	Tyr	Thr	Gln	Ile	Ala	Ile	Arg	Phe	Ala	Glu
2255						2260					2265			
Leu	His	Asp	Thr	Ser	Leu	Arg	Met	Ala	Ala	Lys	Gly	Val	Ile	Lys
2270						2275					2280			
Lys	Val	Val	Asp	Trp	Glu	Glu	Ser	Arg	Ser	Phe	Phe	Tyr	Lys	Arg
2285						2290					2295			
Leu	Arg	Arg	Arg	Ile	Ser	Glu	Asp	Val	Leu	Ala	Lys	Glu	Ile	Arg
2300						2305					2310			
Ala	Val	Ala	Gly	Glu	Gln	Phe	Ser	His	Gln	Pro	Ala	Ile	Glu	Leu
2315						2320					2325			
Ile	Lys	Lys	Trp	Tyr	Ser	Ala	Ser	His	Ala	Ala	Glu	Trp	Asp	Asp
2330						2335					2340			
Asp	Asp	Ala	Phe	Val	Ala	Trp	Met	Asp	Asn	Pro	Glu	Asn	Tyr	Lys
2345						2350					2355			
Asp	Tyr	Ile	Gln	Tyr	Leu	Lys	Ala	Gln	Arg	Val	Ser	Gln	Ser	Leu
2360						2365					2370			

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Ser Ser  Leu Ser Asp Ser Ser  Ser Asp Leu Gln Ala  Leu Pro Gln
2375                2380                2385

Gly Leu  Ser Met Leu Leu Asp  Lys Met Asp Pro Ser  Arg Arg Ala
2390                2395                2400

Gln Leu  Val Glu Glu Ile Arg  Lys Val Leu Gly
2405                2410

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<210> SEQ ID NO 3
<211> LENGTH: 2414
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: mutant ACC

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<400> SEQUENCE: 3

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Met Pro Met Arg Pro Trp Glu Phe Ile Ile Leu Gly Arg Thr Ala Pro
1              5              10              15

Ile Ser Pro Pro Pro Leu Gly Ile Ser Ala Phe Ser Arg Pro Leu Arg
                20              25              30

Leu Arg Leu Arg Leu Arg Leu Ala Ala Val Val Asp Ala Ala Arg Ile
35              40              45

Gln Thr Pro Pro Arg Val Ala Phe Ser Ala Phe Phe Phe Phe Phe
50              55              60

Phe Phe Ser Glu Gly Leu Ser Arg Glu Gly Thr Ala Ala Arg Leu Pro
65              70              75              80

Ala Leu Ile Tyr Arg Val Pro Phe Ile Pro Cys Leu Leu Gln Leu Gln
85              90              95

Ile Arg Ala Pro His Ala Asp Pro Arg Pro Ala Gly Phe Ser Cys Ala
100             105             110

His His Phe Leu Ala Leu Ala Gln Gly Leu Ser Ser Ser Leu Leu Ser
115             120             125

Gly Thr Ile Ser His Ile Trp Gly Phe Ile Phe Ser Leu Tyr Gly Thr
130             135             140

Thr His Leu Arg Asn Arg Ala Ile Leu Leu Phe Gly Leu Ala Gly Ile
145             150             155             160

Ile Asp Leu Pro Asn Asp Ala Ala Ser Glu Val Asp Ile Ser His Gly
165             170             175

Ser Glu Asp Pro Arg Gly Pro Thr Val Pro Gly Ser Tyr Gln Met Asn
180             185             190

Gly Ile Ile Asn Glu Thr His Asn Gly Arg His Ala Ser Val Ser Lys
195             200             205

Val Val Glu Phe Cys Thr Ala Leu Gly Gly Lys Thr Pro Ile His Ser
210             215             220

Val Leu Val Ala Asn Asn Gly Met Ala Ala Ala Lys Phe Met Arg Ser
225             230             235             240

Val Arg Thr Trp Ala Asn Asp Thr Phe Gly Ser Glu Lys Ala Ile Gln
245             250             255

Leu Ile Ala Met Ala Thr Pro Glu Asp Leu Arg Ile Asn Ala Glu His
260             265             270

Ile Arg Ile Ala Asp Gln Phe Val Glu Val Pro Gly Gly Thr Asn Asn
275             280             285

Asn Asn Tyr Ala Asn Val Gln Leu Ile Val Glu Ile Ala Glu Arg Thr
290             295             300

Gly Val Ser Ala Val Trp Pro Gly Trp Gly His Ala Ser Glu Asn Pro
305             310             315             320

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Glu	Leu	Pro	Asp	Ala	Leu	Thr	Ala	Lys	Gly	Ile	Val	Phe	Leu	Gly	Pro	325	330	335
Pro	Ala	Ser	Ser	Met	His	Ala	Leu	Gly	Asp	Lys	Val	Gly	Ser	Ala	Leu	340	345	350
Ile	Ala	Gln	Ala	Ala	Gly	Val	Pro	Thr	Leu	Ala	Trp	Ser	Gly	Ser	His	355	360	365
Val	Glu	Val	Pro	Leu	Glu	Cys	Cys	Leu	Asp	Ser	Ile	Pro	Asp	Glu	Met	370	375	380
Tyr	Arg	Lys	Ala	Cys	Val	Thr	Thr	Thr	Glu	Glu	Ala	Val	Ala	Ser	Cys	385	390	395
Gln	Val	Val	Gly	Tyr	Pro	Ala	Met	Ile	Lys	Ala	Ser	Trp	Gly	Gly	Gly	405	410	415
Gly	Lys	Gly	Ile	Arg	Lys	Val	His	Asn	Asp	Asp	Glu	Val	Arg	Thr	Leu	420	425	430
Phe	Lys	Gln	Val	Gln	Gly	Glu	Val	Pro	Gly	Ser	Pro	Ile	Phe	Ile	Met	435	440	445
Arg	Leu	Ala	Ala	Gln	Ser	Arg	His	Leu	Glu	Val	Gln	Leu	Leu	Cys	Asp	450	455	460
Gln	Tyr	Gly	Asn	Val	Ala	Ala	Leu	His	Ser	Arg	Asp	Cys	Ser	Val	Gln	465	470	475
Arg	Arg	His	Gln	Lys	Ile	Ile	Glu	Glu	Gly	Pro	Val	Thr	Val	Ala	Pro	485	490	495
Arg	Glu	Thr	Val	Lys	Glu	Leu	Glu	Gln	Ala	Ala	Arg	Arg	Leu	Ala	Lys	500	505	510
Ala	Val	Gly	Tyr	Val	Gly	Ala	Ala	Thr	Val	Glu	Tyr	Leu	Tyr	Ser	Met	515	520	525
Glu	Thr	Gly	Glu	Tyr	Tyr	Phe	Leu	Glu	Leu	Asn	Pro	Arg	Leu	Gln	Val	530	535	540
Glu	His	Pro	Val	Thr	Glu	Trp	Ile	Ala	Glu	Val	Asn	Leu	Pro	Ala	Ala	545	550	555
Gln	Val	Ala	Val	Gly	Met	Gly	Ile	Pro	Leu	Trp	Gln	Ile	Pro	Glu	Ile	565	570	575
Arg	Arg	Phe	Tyr	Gly	Met	Asn	His	Gly	Gly	Gly	Tyr	Asp	Leu	Trp	Arg	580	585	590
Lys	Thr	Ala	Ala	Leu	Ala	Thr	Pro	Phe	Asn	Phe	Asp	Glu	Val	Asp	Ser	595	600	605
Lys	Trp	Pro	Lys	Gly	His	Cys	Val	Ala	Val	Arg	Ile	Thr	Ser	Glu	Asp	610	615	620
Pro	Asp	Asp	Gly	Phe	Lys	Pro	Thr	Gly	Gly	Lys	Val	Lys	Glu	Ile	Ser	625	630	635
Phe	Lys	Ser	Lys	Pro	Asn	Val	Trp	Ala	Tyr	Phe	Ser	Val	Lys	Ser	Gly	645	650	655
Gly	Gly	Ile	His	Glu	Phe	Ala	Asp	Ser	Gln	Phe	Gly	His	Val	Phe	Ala	660	665	670
Tyr	Gly	Thr	Thr	Arg	Ser	Ala	Ala	Ile	Thr	Thr	Met	Ala	Leu	Ala	Leu	675	680	685
Lys	Glu	Val	Gln	Ile	Arg	Gly	Glu	Ile	His	Ser	Asn	Val	Asp	Tyr	Thr	690	695	700
Val	Asp	Leu	Leu	Asn	Ala	Ser	Asp	Phe	Arg	Glu	Asn	Lys	Ile	His	Thr	705	710	715
Gly	Trp	Leu	Asp	Thr	Arg	Ile	Ala	Met	Arg	Val	Gln	Ala	Glu	Arg	Pro	725	730	735
Pro	Trp	Tyr	Ile	Ser	Val	Val	Gly	Gly	Ala	Leu	Tyr	Lys	Thr	Val	Thr			

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740						745					750				
Ala	Asn	Thr	Ala	Thr	Val	Ser	Asp	Tyr	Val	Gly	Tyr	Leu	Thr	Lys	Gly
	755						760					765			
Gln	Ile	Pro	Pro	Lys	His	Ile	Ser	Leu	Val	Tyr	Thr	Thr	Val	Ala	Leu
	770					775					780				
Asn	Ile	Asp	Gly	Lys	Lys	Tyr	Thr	Ile	Asp	Thr	Val	Arg	Ser	Gly	His
785					790					795					800
Gly	Ser	Tyr	Arg	Leu	Arg	Met	Asn	Gly	Ser	Thr	Val	Asp	Ala	Asn	Val
				805					810					815	
Gln	Ile	Leu	Cys	Asp	Gly	Gly	Leu	Leu	Met	Gln	Leu	Asp	Gly	Asn	Ser
			820					825					830		
His	Val	Ile	Tyr	Ala	Glu	Glu	Glu	Ala	Ser	Gly	Thr	Arg	Leu	Leu	Ile
	835						840					845			
Asp	Gly	Lys	Thr	Cys	Met	Leu	Gln	Asn	Asp	His	Asp	Pro	Ser	Lys	Leu
	850					855					860				
Leu	Ala	Glu	Thr	Pro	Cys	Lys	Leu	Leu	Arg	Phe	Leu	Val	Ala	Asp	Gly
865					870					875					880
Ala	His	Val	Asp	Ala	Asp	Val	Pro	Tyr	Ala	Glu	Val	Glu	Val	Met	Lys
				885					890					895	
Met	Cys	Met	Pro	Leu	Leu	Ser	Pro	Ala	Ser	Gly	Val	Ile	His	Val	Val
			900					905					910		
Met	Ser	Glu	Gly	Gln	Ala	Met	Gln	Ala	Gly	Asp	Leu	Ile	Ala	Arg	Leu
	915						920					925			
Asp	Leu	Asp	Asp	Pro	Ser	Ala	Val	Lys	Arg	Ala	Glu	Pro	Phe	Glu	Asp
	930					935					940				
Thr	Phe	Pro	Gln	Met	Gly	Leu	Pro	Ile	Ala	Ala	Ser	Gly	Gln	Val	His
945					950					955					960
Lys	Leu	Cys	Ala	Ala	Ser	Leu	Asn	Ala	Cys	Arg	Met	Ile	Leu	Ala	Gly
			965						970					975	
Tyr	Glu	His	Asp	Ile	Asp	Lys	Val	Val	Pro	Glu	Leu	Val	Tyr	Cys	Leu
		980						985					990		
Asp	Thr	Pro	Glu	Leu	Pro	Phe	Leu	Gln	Trp	Glu	Glu	Leu	Met	Ser	Val
	995						1000					1005			
Leu	Ala	Thr	Arg	Leu	Pro	Arg	Asn	Leu	Lys	Ser	Glu	Leu	Glu	Gly	
	1010					1015					1020				
Lys	Tyr	Glu	Glu	Tyr	Lys	Val	Lys	Phe	Asp	Ser	Gly	Ile	Ile	Asn	
	1025					1030					1035				
Asp	Phe	Pro	Ala	Asn	Met	Leu	Arg	Val	Ile	Ile	Glu	Glu	Asn	Leu	
	1040					1045					1050				
Ala	Cys	Gly	Ser	Glu	Lys	Glu	Lys	Ala	Thr	Asn	Glu	Arg	Leu	Val	
	1055					1060					1065				
Glu	Pro	Leu	Met	Ser	Leu	Leu	Lys	Ser	Tyr	Glu	Gly	Gly	Arg	Glu	
	1070					1075					1080				
Ser	His	Ala	His	Phe	Val	Val	Lys	Ser	Leu	Phe	Glu	Glu	Tyr	Leu	
	1085					1090					1095				
Tyr	Val	Glu	Glu	Leu	Phe	Ser	Asp	Gly	Ile	Gln	Ser	Asp	Val	Ile	
	1100					1105					1110				
Glu	Arg	Leu	Arg	Leu	Gln	His	Ser	Lys	Asp	Leu	Gln	Lys	Val	Val	
	1115					1120					1125				
Asp	Ile	Val	Leu	Ser	His	Gln	Ser	Val	Arg	Asn	Lys	Thr	Lys	Leu	
	1130					1135					1140				
Ile	Leu	Lys	Leu	Met	Glu	Ser	Leu	Val	Tyr	Pro	Asn	Pro	Ala	Ala	
	1145					1150					1155				

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Tyr Arg	Asp Gln Leu Ile Arg	Phe Ser Ser Leu Asn	His Lys Ala
1160	1165	1170	
Tyr Tyr	Lys Leu Ala Leu Lys	Ala Ser Glu Leu Leu	Glu Gln Thr
1175	1180	1185	
Lys Leu	Ser Glu Leu Arg Ala	Arg Ile Ala Arg Ser	Leu Ser Glu
1190	1195	1200	
Leu Glu	Met Phe Thr Glu Glu	Ser Lys Gly Leu Ser	Met His Lys
1205	1210	1215	
Arg Glu	Ile Ala Ile Lys Glu	Ser Met Glu Asp Leu	Val Thr Ala
1220	1225	1230	
Pro Leu	Pro Val Glu Asp Ala	Leu Ile Ser Leu Phe	Asp Cys Ser
1235	1240	1245	
Asp Thr	Thr Val Gln Gln Arg	Val Ile Glu Thr Tyr	Ile Ala Arg
1250	1255	1260	
Leu Tyr	Gln Pro His Leu Val	Lys Asp Ser Ile Lys	Met Lys Trp
1265	1270	1275	
Ile Glu	Ser Gly Val Ile Ala	Leu Trp Glu Phe Pro	Glu Gly His
1280	1285	1290	
Phe Asp	Ala Arg Asn Gly Gly	Ala Val Leu Gly Asp	Lys Arg Trp
1295	1300	1305	
Gly Ala	Met Val Ile Val Lys	Ser Leu Glu Ser Leu	Ser Met Ala
1310	1315	1320	
Ile Arg	Phe Ala Leu Lys Glu	Thr Ser His Tyr Thr	Ser Ser Glu
1325	1330	1335	
Gly Asn	Met Met His Ile Ala	Leu Leu Gly Ala Asp	Asn Lys Met
1340	1345	1350	
His Ile	Ile Gln Glu Ser Gly	Asp Asp Ala Asp Arg	Ile Ala Lys
1355	1360	1365	
Leu Pro	Leu Ile Leu Lys Asp	Asn Val Thr Asp Leu	His Ala Ser
1370	1375	1380	
Gly Val	Lys Thr Ile Ser Phe	Ile Val Gln Arg Asp	Glu Ala Arg
1385	1390	1395	
Met Thr	Met Arg Arg Thr Phe	Leu Trp Ser Asp Glu	Lys Leu Ser
1400	1405	1410	
Tyr Glu	Glu Glu Pro Ile Leu	Arg His Val Glu Pro	Pro Leu Ser
1415	1420	1425	
Ala Leu	Leu Glu Leu Asp Lys	Leu Lys Val Lys Gly	Tyr Asn Glu
1430	1435	1440	
Met Lys	Tyr Thr Pro Ser Arg	Asp Arg Gln Trp His	Ile Tyr Thr
1445	1450	1455	
Leu Arg	Asn Thr Glu Asn Pro	Lys Met Leu His Arg	Val Phe Phe
1460	1465	1470	
Arg Thr	Leu Val Arg Gln Pro	Ser Val Ser Asn Lys	Phe Ser Ser
1475	1480	1485	
Gly Gln	Ile Gly Asp Met Glu	Val Gly Ser Ala Glu	Glu Pro Leu
1490	1495	1500	
Ser Phe	Thr Ser Thr Ser Ile	Leu Arg Ser Leu Met	Thr Ala Ile
1505	1510	1515	
Glu Glu	Leu Glu Leu His Ala	Ile Arg Thr Gly His	Ser His Met
1520	1525	1530	
Tyr Leu	His Val Leu Lys Glu	Gln Lys Leu Leu Asp	Leu Val Pro
1535	1540	1545	

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Val	Ser	Gly	Asn	Thr	Val	Leu	Asp	Val	Gly	Gln	Asp	Glu	Ala	Thr
1550						1555					1560			
Ala	Tyr	Ser	Leu	Leu	Lys	Glu	Met	Ala	Met	Lys	Ile	His	Glu	Leu
1565						1570					1575			
Val	Gly	Ala	Arg	Met	His	His	Leu	Ser	Val	Cys	Gln	Trp	Glu	Val
1580						1585					1590			
Lys	Leu	Lys	Leu	Asp	Cys	Asp	Gly	Pro	Ala	Ser	Gly	Thr	Trp	Arg
1595						1600					1605			
Ile	Val	Thr	Thr	Asn	Val	Thr	Ser	His	Thr	Cys	Thr	Val	Asp	Ile
1610						1615					1620			
Tyr	Arg	Glu	Met	Glu	Asp	Lys	Glu	Ser	Arg	Lys	Leu	Val	Tyr	His
1625						1630					1635			
Pro	Ala	Thr	Pro	Ala	Ala	Gly	Pro	Leu	His	Gly	Val	Ala	Leu	Asn
1640						1645					1650			
Asn	Pro	Tyr	Gln	Pro	Leu	Ser	Val	Ile	Asp	Leu	Lys	Arg	Cys	Ser
1655						1660					1665			
Ala	Arg	Asn	Asn	Arg	Thr	Thr	Tyr	Cys	Tyr	Asp	Phe	Pro	Leu	Ala
1670						1675					1680			
Phe	Glu	Thr	Ala	Val	Arg	Lys	Ser	Trp	Ser	Ser	Ser	Thr	Ser	Gly
1685						1690					1695			
Ala	Ser	Lys	Gly	Val	Glu	Asn	Ala	Gln	Cys	Tyr	Val	Lys	Ala	Thr
1700						1705					1710			
Glu	Leu	Val	Phe	Ala	Asp	Lys	His	Gly	Ser	Trp	Gly	Thr	Pro	Leu
1715						1720					1725			
Val	Gln	Met	Asp	Arg	Pro	Ala	Gly	Leu	Asn	Asp	Ile	Gly	Met	Val
1730						1735					1740			
Ala	Trp	Thr	Leu	Lys	Met	Ser	Thr	Pro	Glu	Phe	Pro	Ser	Gly	Arg
1745						1750					1755			
Glu	Ile	Ile	Val	Val	Ala	Asn	Asp	Ile	Thr	Phe	Arg	Ala	Gly	Ser
1760						1765					1770			
Phe	Gly	Pro	Arg	Glu	Asp	Ala	Phe	Phe	Glu	Ala	Val	Thr	Asn	Leu
1775						1780					1785			
Ala	Cys	Glu	Lys	Lys	Leu	Pro	Leu	Ile	Tyr	Leu	Ala	Ala	Asn	Ser
1790						1795					1800			
Gly	Ala	Arg	Ile	Gly	Ile	Ala	Asp	Glu	Val	Lys	Ser	Cys	Phe	Arg
1805						1810					1815			
Val	Gly	Trp	Ser	Asp	Asp	Gly	Ser	Pro	Glu	Arg	Gly	Phe	Gln	Tyr
1820						1825					1830			
Ile	Tyr	Leu	Ser	Glu	Glu	Asp	Tyr	Ala	Arg	Ile	Gly	Thr	Ser	Val
1835						1840					1845			
Ile	Ala	His	Lys	Met	Gln	Leu	Asp	Ser	Gly	Glu	Ile	Arg	Trp	Val
1850						1855					1860			
Ile	Asp	Ser	Val	Val	Gly	Lys	Glu	Asp	Gly	Leu	Gly	Val	Glu	Asn
1865						1870					1875			
Ile	His	Gly	Ser	Ala	Ala	Ile	Ala	Ser	Ala	Tyr	Ser	Arg	Ala	Tyr
1880						1885					1890			
Lys	Glu	Thr	Phe	Thr	Leu	Thr	Phe	Val	Thr	Gly	Arg	Thr	Val	Gly
1895						1900					1905			
Ile	Gly	Ala	Tyr	Leu	Ala	Arg	Leu	Gly	Ile	Arg	Cys	Ile	Gln	Arg
1910						1915					1920			
Leu	Asp	Gln	Pro	Ile	Ile	Leu	Thr	Gly	Tyr	Ser	Ala	Leu	Asn	Lys
1925						1930					1935			
Leu	Leu	Gly	Arg	Glu	Val	Tyr	Ser	Ser	His	Met	Gln	Leu	Gly	Gly

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1940	1945	1950
Pro Lys Ile Met Ala Thr	Asn Gly Val Val His	Leu Thr Val Ser
1955	1960	1965
Asp Asp Leu Glu Gly Val	Ser Asn Ile Leu Arg	Trp Leu Ser Tyr
1970	1975	1980
Val Pro Ala Tyr Ile Gly	Gly Pro Leu Pro Val Thr	Thr Pro Leu
1985	1990	1995
Asp Pro Pro Asp Arg Pro	Val Ala Tyr Ile Pro	Glu Asn Ser Cys
2000	2005	2010
Asp Pro Arg Ala Ala Ile	Arg Gly Val Asp Asp	Ser Gln Gly Lys
2015	2020	2025
Trp Leu Gly Gly Met Phe	Asp Lys Asp Ser Phe	Val Glu Thr Phe
2030	2035	2040
Glu Gly Trp Ala Lys Thr	Val Val Thr Gly Arg	Ala Lys Leu Gly
2045	2050	2055
Gly Ile Pro Val Gly Val	Ile Ala Val Glu Thr	Gln Thr Met Met
2060	2065	2070
Gln Thr Ile Pro Ala Asp	Pro Gly Gln Leu Asp	Ser Arg Glu Gln
2075	2080	2085
Ser Val Pro Arg Ala Gly	Gln Val Trp Phe Pro	Asp Ser Ala Thr
2090	2095	2100
Lys Thr Ala Gln Ala Leu	Leu Asp Phe Asn Arg	Glu Gly Leu Pro
2105	2110	2115
Leu Phe Ile Leu Ala Asn	Trp Arg Gly Phe Ser	Gly Gly Gln Arg
2120	2125	2130
Asp Leu Phe Glu Gly Ile	Leu Gln Ala Gly Ser	Thr Ile Val Glu
2135	2140	2145
Asn Leu Arg Thr Tyr Asn	Gln Pro Ala Phe Val	Tyr Ile Pro Met
2150	2155	2160
Ala Ala Glu Leu Arg Gly	Gly Ala Trp Val Val	Val Asp Ser Lys
2165	2170	2175
Ile Asn Pro Asp Arg Ile	Glu Arg Tyr Ala Glu	Arg Thr Ala Lys
2180	2185	2190
Gly Asn Val Leu Glu Pro	Gln Gly Leu Ile Glu	Ile Lys Phe Arg
2195	2200	2205
Ser Glu Glu Leu Gln Asp	Cys Met Ser Arg Leu	Asp Pro Thr Leu
2210	2215	2220
Ile Asp Leu Lys Ala Lys	Leu Glu Val Ala Asn	Lys Asn Gly Ser
2225	2230	2235
Ala Asp Thr Lys Ser Leu	Gln Glu Asn Ile Glu	Ala Arg Thr Lys
2240	2245	2250
Gln Leu Met Pro Leu Tyr	Thr Gln Ile Ala Ile	Arg Phe Ala Glu
2255	2260	2265
Leu His Asp Thr Ser Leu	Arg Met Ala Ala Lys	Gly Val Ile Lys
2270	2275	2280
Lys Val Val Asp Trp Glu	Glu Ser Arg Ser Phe	Phe Tyr Lys Arg
2285	2290	2295
Leu Arg Arg Arg Ile Ser	Glu Asp Val Leu Ala	Lys Glu Ile Arg
2300	2305	2310
Ala Val Ala Gly Glu Gln	Phe Ser His Gln Pro	Ala Ile Glu Leu
2315	2320	2325
Ile Lys Lys Trp Tyr Ser	Ala Ser His Ala Ala	Glu Trp Asp Asp
2330	2335	2340

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Asp Asp Ala Phe Val Ala Trp Met Asp Asn Pro Glu Asn Tyr Lys
 2345 2350 2355
 Asp Tyr Ile Gln Tyr Leu Lys Ala Gln Arg Val Ser Gln Ser Leu
 2360 2365 2370
 Ser Ser Leu Ser Asp Ser Ser Ser Asp Leu Gln Ala Leu Pro Gln
 2375 2380 2385
 Gly Leu Ser Met Leu Leu Asp Lys Met Asp Pro Ser Arg Arg Ala
 2390 2395 2400
 Gln Leu Val Glu Glu Ile Arg Lys Val Leu Gly
 2405 2410

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 <211> LENGTH: 2414
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: mutant ACC

<400> SEQUENCE: 4

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 20 25 30
 Leu Arg Leu Arg Leu Arg Leu Ala Ala Val Val Asp Ala Ala Arg Ile
 35 40 45
 Gln Thr Pro Pro Arg Val Ala Phe Ser Ala Phe Phe Phe Phe Phe
 50 55 60
 Phe Phe Ser Glu Gly Leu Ser Arg Glu Gly Thr Ala Ala Arg Leu Pro
 65 70 75 80
 Ala Leu Ile Tyr Arg Val Pro Phe Ile Pro Cys Leu Leu Gln Leu Gln
 85 90 95
 Ile Arg Ala Pro His Ala Asp Pro Arg Pro Ala Gly Phe Ser Cys Ala
 100 105 110
 His His Phe Leu Ala Leu Ala Gln Gly Leu Ser Ser Ser Leu Leu Ser
 115 120 125
 Gly Thr Ile Ser His Ile Trp Gly Phe Ile Phe Ser Leu Tyr Gly Thr
 130 135 140
 Thr His Leu Arg Asn Arg Ala Ile Leu Leu Phe Gly Leu Ala Gly Ile
 145 150 155 160
 Ile Asp Leu Pro Asn Asp Ala Ala Ser Glu Val Asp Ile Ser His Gly
 165 170 175
 Ser Glu Asp Pro Arg Gly Pro Thr Val Pro Gly Ser Tyr Gln Met Asn
 180 185 190
 Gly Ile Ile Asn Glu Thr His Asn Gly Arg His Ala Ser Val Ser Lys
 195 200 205
 Val Val Glu Phe Cys Thr Ala Leu Gly Gly Lys Thr Pro Ile His Ser
 210 215 220
 Val Leu Val Ala Asn Asn Gly Met Ala Ala Ala Lys Phe Met Arg Ser
 225 230 235 240
 Val Arg Thr Trp Ala Asn Asp Thr Phe Gly Ser Glu Lys Ala Ile Gln
 245 250 255
 Leu Ile Ala Met Ala Thr Pro Glu Asp Leu Arg Ile Asn Ala Glu His
 260 265 270
 Ile Arg Ile Ala Asp Gln Phe Val Glu Val Pro Gly Gly Thr Asn Asn
 275 280 285

Asn	Asn	Tyr	Ala	Asn	Val	Gln	Leu	Ile	Val	Glu	Ile	Ala	Glu	Arg	Thr
290						295				300					
Gly	Val	Ser	Ala	Val	Trp	Pro	Gly	Trp	Gly	His	Ala	Ser	Glu	Asn	Pro
305				310						315				320	
Glu	Leu	Pro	Asp	Ala	Leu	Thr	Ala	Lys	Gly	Ile	Val	Phe	Leu	Gly	Pro
				325				330						335	
Pro	Ala	Ser	Ser	Met	His	Ala	Leu	Gly	Asp	Lys	Val	Gly	Ser	Ala	Leu
		340						345				350			
Ile	Ala	Gln	Ala	Ala	Gly	Val	Pro	Thr	Leu	Ala	Trp	Ser	Gly	Ser	His
355						360						365			
Val	Glu	Val	Pro	Leu	Glu	Cys	Cys	Leu	Asp	Ser	Ile	Pro	Asp	Glu	Met
370						375				380					
Tyr	Arg	Lys	Ala	Cys	Val	Thr	Thr	Thr	Glu	Glu	Ala	Val	Ala	Ser	Cys
385				390						395				400	
Gln	Val	Val	Gly	Tyr	Pro	Ala	Met	Ile	Lys	Ala	Ser	Trp	Gly	Gly	Gly
				405				410						415	
Gly	Lys	Gly	Ile	Arg	Lys	Val	His	Asn	Asp	Asp	Glu	Val	Arg	Thr	Leu
				420				425				430			
Phe	Lys	Gln	Val	Gln	Gly	Glu	Val	Pro	Gly	Ser	Pro	Ile	Phe	Ile	Met
435						440						445			
Arg	Leu	Ala	Ala	Gln	Ser	Arg	His	Leu	Glu	Val	Gln	Leu	Leu	Cys	Asp
450						455				460					
Gln	Tyr	Gly	Asn	Val	Ala	Ala	Leu	His	Ser	Arg	Asp	Cys	Ser	Val	Gln
465				470						475				480	
Arg	Arg	His	Gln	Lys	Ile	Ile	Glu	Glu	Gly	Pro	Val	Thr	Val	Ala	Pro
				485				490						495	
Arg	Glu	Thr	Val	Lys	Glu	Leu	Glu	Gln	Ala	Ala	Arg	Arg	Leu	Ala	Lys
		500				505						510			
Ala	Val	Gly	Tyr	Val	Gly	Ala	Ala	Thr	Val	Glu	Tyr	Leu	Tyr	Ser	Met
515						520				525					
Glu	Thr	Gly	Glu	Tyr	Tyr	Phe	Leu	Glu	Leu	Asn	Pro	Arg	Leu	Gln	Val
530						535				540					
Glu	His	Pro	Val	Thr	Glu	Trp	Ile	Ala	Glu	Val	Asn	Leu	Pro	Ala	Ala
545				550						555				560	
Gln	Val	Ala	Val	Gly	Met	Gly	Ile	Pro	Leu	Trp	Gln	Ile	Pro	Glu	Ile
				565				570						575	
Arg	Arg	Phe	Tyr	Gly	Met	Asn	His	Gly	Gly	Gly	Tyr	Asp	Leu	Trp	Arg
		580						585				590			
Lys	Thr	Ala	Ala	Leu	Ala	Thr	Pro	Phe	Asn	Phe	Asp	Glu	Val	Asp	Ser
595						600				605					
Lys	Trp	Pro	Lys	Gly	His	Cys	Val	Ala	Val	Arg	Ile	Thr	Ser	Glu	Asp
610						615				620					
Pro	Asp	Asp	Gly	Phe	Lys	Pro	Thr	Gly	Gly	Lys	Val	Lys	Glu	Ile	Ser
625				630						635				640	
Phe	Lys	Ser	Lys	Pro	Asn	Val	Trp	Ala	Tyr	Phe	Ser	Val	Lys	Ser	Gly
				645				650						655	
Gly	Gly	Ile													

Val 705	Asp	Leu	Leu	Asn	Ala	Ser	Asp	Phe	Arg	Glu	Asn	Lys	Ile	His	Thr
				710						715					
Gly	Trp	Leu	Asp	Thr	Arg	Ile	Ala	Met	Arg	Val	Gln	Ala	Glu	Arg	Pro
				725						730					
Pro	Trp	Tyr	Ile	Ser	Val	Val	Gly	Gly	Ala	Leu	Tyr	Lys	Thr	Val	Thr
				740						745					
Ala	Asn	Thr	Ala	Thr	Val	Ser	Asp	Tyr	Val	Gly	Tyr	Leu	Thr	Lys	Gly
				755						760					
Gln	Ile	Pro	Pro	Lys	His	Ile	Ser	Leu	Val	Tyr	Thr	Thr	Val	Ala	Leu
				770						780					
Asn	Ile	Asp	Gly	Lys	Lys	Tyr	Thr	Ile	Asp	Thr	Val	Arg	Ser	Gly	His
				785						795					
Gly	Ser	Tyr	Arg	Leu	Arg	Met	Asn	Gly	Ser	Thr	Val	Asp	Ala	Asn	Val
				805						810					
Gln	Ile	Leu	Cys	Asp	Gly	Gly	Leu	Leu	Met	Gln	Leu	Asp	Gly	Asn	Ser
				820						825					
His	Val	Ile	Tyr	Ala	Glu	Glu	Glu	Ala	Ser	Gly	Thr	Arg	Leu	Leu	Ile
				835						840					
Asp	Gly	Lys	Thr	Cys	Met	Leu	Gln	Asn	Asp	His	Asp	Pro	Ser	Lys	Leu
				850						860					
Leu	Ala	Glu	Thr	Pro	Cys	Lys	Leu	Leu	Arg	Phe	Leu	Val	Ala	Asp	Gly
				865						875					
Ala	His	Val	Asp	Ala	Asp	Val	Pro	Tyr	Ala	Glu	Val	Glu	Val	Met	Lys
				885						890					
Met	Cys	Met	Pro	Leu	Leu	Ser	Pro	Ala	Ser	Gly	Val	Ile	His	Val	Val
				900						905					
Met	Ser	Glu	Gly	Gln	Ala	Met	Gln	Ala	Gly	Asp	Leu	Ile	Ala	Arg	Leu
				915						920					
Asp	Leu	Asp	Asp	Pro	Ser	Ala	Val	Lys	Arg	Ala	Glu	Pro	Phe	Glu	Asp
				930						940					
Thr	Phe	Pro	Gln	Met	Gly	Leu	Pro	Ile	Ala	Ala	Ser	Gly	Gln	Val	His
				945						955					
Lys	Leu	Cys	Ala	Ala	Ser	Leu	Asn	Ala	Cys	Arg	Met	Ile	Leu	Ala	Gly
				965						970					
Tyr	Glu	His	Asp	Ile	Asp	Lys	Val	Val	Pro	Glu	Leu	Val	Tyr	Cys	Leu
				980						985					
Asp	Thr	Pro	Glu	Leu	Pro	Phe	Leu	Gln	Trp	Glu	Glu	Leu	Met	Ser	Val
				995						1000					
Leu	Ala	Thr	Arg	Leu	Pro	Arg	Asn	Leu	Lys	Ser	Glu	Leu	Glu	Gly	
				1010						1015					
Lys	Tyr	Glu	Glu	Tyr	Lys	Val	Lys	Phe	Asp	Ser	Gly	Ile	Ile	Asn	
				1025						1030					
Asp	Phe	Pro	Ala	Asn	Met	Leu	Arg	Val	Ile	Ile	Glu	Glu	Asn	Leu	
				1040						1045					
Ala	Cys	Gly	Ser	Glu	Lys	Glu	Lys	Ala	Thr	Asn	Glu	Arg	Leu	Val	
				1055						1060					
Glu	Pro	Leu	Met	Ser	Leu	Leu	Lys	Ser	Tyr	Glu	Gly	Gly	Arg	Glu	
				1070						1075					
Ser	His	Ala	His	Phe	Val	Val	Lys	Ser	Leu	Phe	Glu	Glu	Tyr	Leu	
				1085						1090					
Tyr	Val	Glu	Glu	Leu	Phe	Ser	Asp	Gly	Ile	Gln	Ser	Asp	Val	Ile	
				1100						1105					
Glu	Arg	Leu	Arg	Leu	Gln	His	Ser	Lys	Asp	Leu	Gln	Lys	Val	Val	

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1115	1120	1125
Asp Ile Val Leu Ser His Gln Ser Val Arg Asn Lys Thr Lys Leu		
1130	1135	1140
Ile Leu Lys Leu Met Glu Ser Leu Val Tyr Pro Asn Pro Ala Ala		
1145	1150	1155
Tyr Arg Asp Gln Leu Ile Arg Phe Ser Ser Leu Asn His Lys Ala		
1160	1165	1170
Tyr Tyr Lys Leu Ala Leu Lys Ala Ser Glu Leu Leu Glu Gln Thr		
1175	1180	1185
Lys Leu Ser Glu Leu Arg Ala Arg Ile Ala Arg Ser Leu Ser Glu		
1190	1195	1200
Leu Glu Met Phe Thr Glu Glu Ser Lys Gly Leu Ser Met His Lys		
1205	1210	1215
Arg Glu Ile Ala Ile Lys Glu Ser Met Glu Asp Leu Val Thr Ala		
1220	1225	1230
Pro Leu Pro Val Glu Asp Ala Leu Ile Ser Leu Phe Asp Cys Ser		
1235	1240	1245
Asp Thr Thr Val Gln Gln Arg Val Ile Glu Thr Tyr Ile Ala Arg		
1250	1255	1260
Leu Tyr Gln Pro His Leu Val Lys Asp Ser Ile Lys Met Lys Trp		
1265	1270	1275
Ile Glu Ser Gly Val Ile Ala Leu Trp Glu Phe Pro Glu Gly His		
1280	1285	1290
Phe Asp Ala Arg Asn Gly Gly Ala Val Leu Gly Asp Lys Arg Trp		
1295	1300	1305
Gly Ala Met Val Ile Val Lys Ser Leu Glu Ser Leu Ser Met Ala		
1310	1315	1320
Ile Arg Phe Ala Leu Lys Glu Thr Ser His Tyr Thr Ser Ser Glu		
1325	1330	1335
Gly Asn Met Met His Ile Ala Leu Leu Gly Ala Asp Asn Lys Met		
1340	1345	1350
His Ile Ile Gln Glu Ser Gly Asp Asp Ala Asp Arg Ile Ala Lys		
1355	1360	1365
Leu Pro Leu Ile Leu Lys Asp Asn Val Thr Asp Leu His Ala Ser		
1370	1375	1380
Gly Val Lys Thr Ile Ser Phe Ile Val Gln Arg Asp Glu Ala Arg		
1385	1390	1395
Met Thr Met Arg Arg Thr Phe Leu Trp Ser Asp Glu Lys Leu Ser		
1400	1405	1410
Tyr Glu Glu Glu Pro Ile Leu Arg His Val Glu Pro Pro Leu Ser		
1415	1420	1425
Ala Leu Leu Glu Leu Asp Lys Leu Lys Val Lys Gly Tyr Asn Glu		
1430	1435	1440
Met Lys Tyr Thr Pro Ser Arg Asp Arg Gln Trp His Ile Tyr Thr		
1445	1450	1455
Leu Arg Asn Thr Glu Asn Pro Lys Met Leu His Arg Val Phe Phe		
1460	1465	1470
Arg Thr Leu Val Arg Gln Pro Ser Val Ser Asn Lys Phe Ser Ser		
1475	1480	1485
Gly Gln Ile Gly Asp Met Glu Val Gly Ser Ala Glu Glu Pro Leu		
1490	1495	1500
Ser Phe Thr Ser Thr Ser Ile Leu Arg Ser Leu Met Thr Ala Ile		
1505	1510	1515

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Glu	Glu	Leu	Glu	Leu	His	Ala	Ile	Arg	Thr	Gly	His	Ser	His	Met
1520						1525					1530			
Tyr	Leu	His	Val	Leu	Lys	Glu	Gln	Lys	Leu	Leu	Asp	Leu	Val	Pro
1535						1540					1545			
Val	Ser	Gly	Asn	Thr	Val	Leu	Asp	Val	Gly	Gln	Asp	Glu	Ala	Thr
1550						1555					1560			
Ala	Tyr	Ser	Leu	Leu	Lys	Glu	Met	Ala	Met	Lys	Ile	His	Glu	Leu
1565						1570					1575			
Val	Gly	Ala	Arg	Met	His	His	Leu	Ser	Val	Cys	Gln	Trp	Glu	Val
1580						1585					1590			
Lys	Leu	Lys	Leu	Asp	Cys	Asp	Gly	Pro	Ala	Ser	Gly	Thr	Trp	Arg
1595						1600					1605			
Ile	Val	Thr	Thr	Asn	Val	Thr	Ser	His	Thr	Cys	Thr	Val	Asp	Ile
1610						1615					1620			
Tyr	Arg	Glu	Met	Glu	Asp	Lys	Glu	Ser	Arg	Lys	Leu	Val	Tyr	His
1625						1630					1635			
Pro	Ala	Thr	Pro	Ala	Ala	Gly	Pro	Leu	His	Gly	Val	Ala	Leu	Asn
1640						1645					1650			
Asn	Pro	Tyr	Gln	Pro	Leu	Ser	Val	Ile	Asp	Leu	Lys	Arg	Cys	Ser
1655						1660					1665			
Ala	Arg	Asn	Asn	Arg	Thr	Thr	Tyr	Cys	Tyr	Asp	Phe	Pro	Leu	Ala
1670						1675					1680			
Phe	Glu	Thr	Ala	Val	Arg	Lys	Ser	Trp	Ser	Ser	Ser	Thr	Ser	Gly
1685						1690					1695			
Ala	Ser	Lys	Gly	Val	Glu	Asn	Ala	Gln	Cys	Tyr	Val	Lys	Ala	Thr
1700						1705					1710			
Glu	Leu	Val	Phe	Ala	Asp	Lys	His	Gly	Ser	Trp	Gly	Thr	Pro	Leu
1715						1720					1725			
Val	Gln	Met	Asp	Arg	Pro	Ala	Gly	Leu	Asn	Asp	Ile	Gly	Met	Val
1730						1735					1740			
Ala	Trp	Thr	Leu	Lys	Met	Ser	Thr	Pro	Glu	Phe	Pro	Ser	Gly	Arg
1745						1750					1755			
Glu	Ile	Ile	Val	Val	Ala	Asn	Asp	Ile	Thr	Phe	Arg	Ala	Gly	Ser
1760						1765					1770			
Phe	Gly	Pro	Arg	Glu	Asp	Ala	Phe	Phe	Glu	Ala	Val	Thr	Asn	Leu
1775						1780					1785			
Ala	Cys	Glu	Lys	Lys	Leu	Pro	Leu	Ile	Tyr	Leu	Ala	Ala	Asn	Ser
1790						1795					1800			
Gly	Ala	Arg	Ile	Gly	Ile	Ala	Asp	Glu	Val	Lys	Ser	Cys	Phe	Arg
1805						1810					1815			
Val	Gly	Trp	Ser	Asp	Asp	Gly	Ser	Pro	Glu	Arg	Gly	Phe	Gln	Tyr
1820						1825					1830			
Ile	Tyr	Leu	Ser	Glu	Glu	Asp	Tyr	Ala	Arg	Ile	Gly	Thr	Ser	Val
1835						1840					1845			
Ile	Ala	His	Lys	Met	Gln	Leu	Asp	Ser	Gly	Glu	Ile	Arg	Trp	Val
1850						1855					1860			
Ile	Asp	Ser	Val	Val	Gly	Lys	Glu	Asp	Gly	Leu	Gly	Val	Glu	Asn
1865						1870					1875			
Val	His	Gly	Ser	Ala	Ala	Ile	Ala	Ser	Ala	Tyr	Ser	Arg	Ala	Tyr
1880						1885					1890			
Lys	Glu	Thr	Phe	Thr	Leu	Thr	Phe	Val	Thr	Gly	Arg	Thr	Val	Gly
1895						1900					1905			

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Ile Gly 1910	Ala Tyr	Leu Ala	Arg 1915	Leu Gly	Ile Arg	Cys 1920	Ile Gln	Arg
Leu Asp 1925	Gln Pro	Ile Ile	Leu 1930	Thr Gly	Tyr Ser	Ala 1935	Leu Asn	Lys
Leu Leu 1940	Gly Arg	Glu Val	Tyr 1945	Ser Ser	His Met	Gln 1950	Leu Gly	Gly
Pro Lys 1955	Ile Met	Ala Thr	Asn 1960	Gly Val	Val His	Leu 1965	Thr Val	Ser
Asp Asp 1970	Leu Glu	Gly Val	Ser 1975	Asn Ile	Leu Arg	Trp 1980	Leu Ser	Tyr
Val Pro 1985	Ala Tyr	Ile Gly	Gly 1990	Pro Leu	Pro Val	Thr 1995	Thr Pro	Leu
Asp Pro 2000	Pro Asp	Arg Pro	Val 2005	Ala Tyr	Ile Pro	Glu 2010	Asn Ser	Cys
Asp Pro 2015	Arg Ala	Ala Ile	Arg 2020	Gly Val	Asp Asp	Ser 2025	Gln Gly	Lys
Trp Leu 2030	Gly Gly	Met Phe	Asp 2035	Lys Asp	Ser Phe	Val 2040	Glu Thr	Phe
Glu Gly 2045	Trp Ala	Lys Thr	Val 2050	Val Thr	Gly Arg	Ala 2055	Lys Leu	Gly
Gly Ile 2060	Pro Val	Gly Val	Ile 2065	Ala Val	Glu Thr	Gln 2070	Thr Met	Met
Gln Thr 2075	Ile Pro	Ala Asp	Pro 2080	Gly Gln	Leu Asp	Ser 2085	Arg Glu	Gln
Ser Val 2090	Pro Arg	Ala Gly	Gln 2095	Val Trp	Phe Pro	Asp 2100	Ser Ala	Thr
Lys Thr 2105	Ala Gln	Ala Leu	Leu 2110	Asp Phe	Asn Arg	Glu 2115	Gly Leu	Pro
Leu Phe 2120	Ile Leu	Ala Asn	Trp 2125	Arg Gly	Phe Ser	Gly 2130	Gly Gln	Arg
Asp Leu 2135	Phe Glu	Gly Ile	Leu 2140	Gln Ala	Gly Ser	Thr 2145	Ile Val	Glu
Asn Leu 2150	Arg Thr	Tyr Asn	Gln 2155	Pro Ala	Phe Val	Tyr 2160	Ile Pro	Met
Ala Ala 2165	Glu Leu	Arg Gly	Gly 2170	Ala Trp	Val Val	Val 2175	Asp Ser	Lys
Ile Asn 2180	Pro Asp	Arg Ile	Glu 2185	Arg Tyr	Ala Glu	Arg 2190	Thr Ala	Lys
Gly Asn 2195	Val Leu	Glu Pro	Gln 2200	Gly Leu	Ile Glu	Ile 2205	Lys Phe	Arg
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What is claimed is:

1. A method for imparting phentodim resistance to a plant, comprising a step of allowing endogenous acetyl-CoA carboxylase (ACC) of the plant to mutate to introduce a mutant ACC into a plant cell, a plant seed, a plant tissue, a plant part, or the plant, wherein compared to an amino acid sequence of a parent ACC, the mutant ACC has a mutation at an amino acid corresponding to amino acid 1,879 of an amino acid sequence shown in SEQ ID NO: 1, wherein the amino acid 1,879 is mutated into valine (V); and the plant is rice;

wherein a method for controlling weeds comprises:

- providing a phentodim-resistant plant prepared by the method for imparting the phentodim resistance to the plant; and
- applying an effective amount of phentodim to the phentodim-resistant plant and the weeds nearby to control the weeds near the phentodim-resistant plant.

2. The method according to claim 1, wherein the parent ACC is derived from a monocotyledonous plant or a dicotyledonous plant.

3. The method according to claim 2, wherein the parent ACC is derived from *Oryza sativa*.

4. The method according to claim 1, wherein the method comprises a step of allowing the mutant ACC to express in the plant cell, the plant seed, the plant tissue, the plant part, or the plant.

5. The method according to claim 1, wherein the method comprises:

introducing the mutant ACC with a gene editing tool.

6. The method according to claim 5, wherein the gene editing tool is one or more selected from the group consisting of clustered regularly interspaced short palindromic repeat (CRISPR), transcription activator-like effector nuclease (TALEN), and zinc-finger nuclease (ZFN).

7. The method according to claim 2, wherein the method comprises a step of allowing the mutant ACC to express in the plant cell, the plant seed, the plant tissue, the plant part, or the plant.

105

8. The method according to claim 3, wherein the method comprises a step of allowing the mutant ACC to express in the plant cell, the plant seed, the plant tissue, the plant part, or the plant.

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106